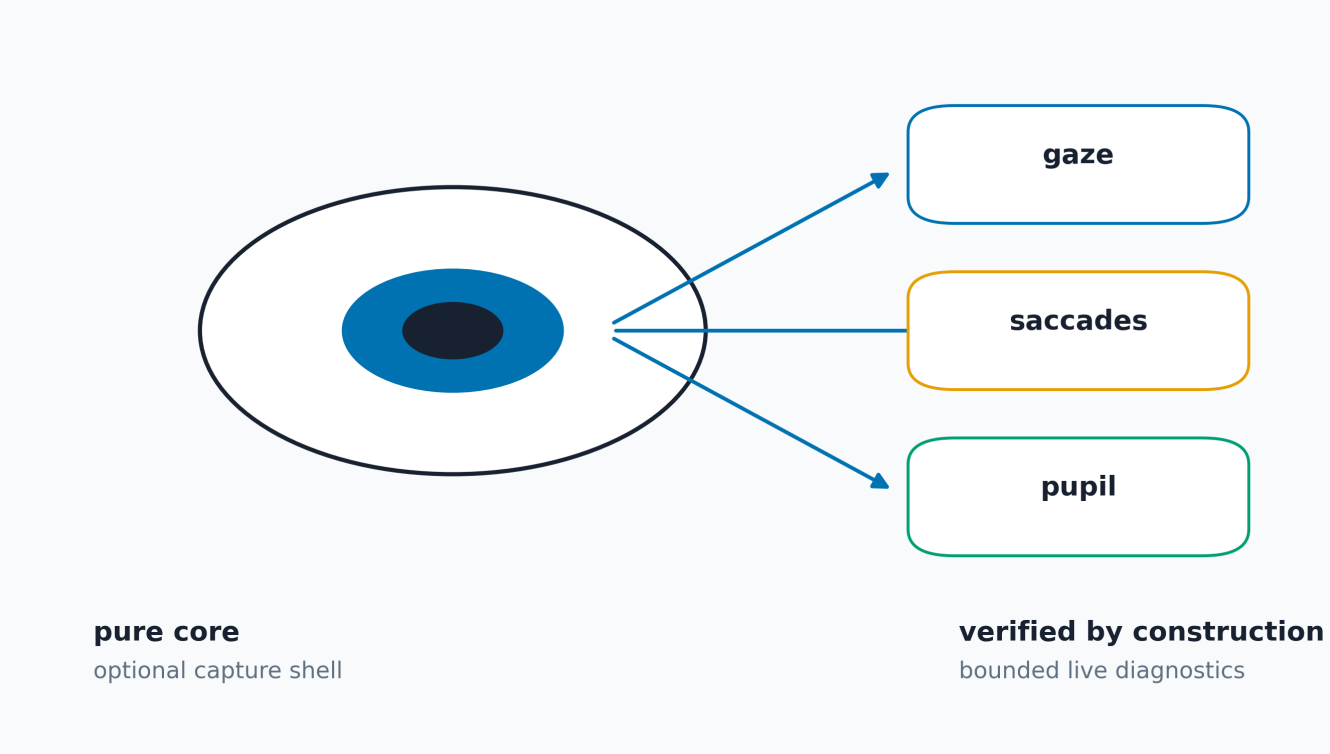
iTrace: an algorithmically verified open-source core for webcam gaze, saccade, and pupil analysis

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Algorithmically verified gaze, saccade, and pupil analysis with a pure core, optional webcam capture, and bounded live empirical diagnostics.

1 Abstract

Webcam eye tracking can make gaze, saccade, and pupil analysis broadly accessible, but commodity-camera software often entangles three concerns that need different evidence: camera capture, scientific signal processing, and claims about real-eye accuracy. We present **iTrace**, a Python toolkit that separates those concerns into (i) a pure NumPy/SciPy core for gaze geometry, Savitzky-Golay velocity, I-VT and I-DT event identification, Engbert-Kliegl microsaccade detection, main-sequence fitting, causal pupil-phase detection, pupil preprocessing, scanpath statistics, and benchmark scoring; (ii) an optional webcam/MediaPipe capture shell; and (iii) bounded diagnostic/export interfaces for live sessions and user-supplied truth files. The core never imports a hardware dependency, so its algorithms are exercised headlessly against constructed ground truth rather than camera availability.

The resulting evidence is algorithmic, explicit, and reproducible. On synthetic traces, the I-VT detector recovers a 10 deg saccade's amplitude to within 5% and peak velocity to within 10%, and an independently formulated reference implementation cross-checks event boundaries. A 3-D eyeball forward model projects known gaze through a pinhole camera to landmarks and back through the estimator, recovering gaze to 0.16 deg RMS; this exercises the geometry path without claiming real-device accuracy. A seeded Monte-Carlo landmark-noise sweep then shows the expected ordering of fragility: gaze remains below a 2 deg RMS bound to sigma approximately 0.0048 (about 3 px at 640 px width), while one-to-one overlap-based saccade recovery falls below F1 = 0.8 near sigma 0.0014 (about 0.92 px). This is event recovery, not a timing-accuracy measurement. The pupil channel is reported conditionally because its noise-sweep robustness follows from the modelling assumption used to inject pupil noise. iTrace also ships a typer CLI, Streamlit dashboard, local HTML/WebSocket orchestrator, guided derived-record empirical workflow, and publication-figure scripts, all under 737 tests at 91.05% coverage. Its contribution is not a claim that webcams are validated eye trackers; it is a verification-first reference implementation whose evidence boundaries are visible in the code, figures, reports, and manuscript. iTrace is MIT-licensed and openly released at https://github.com/docxology/itrace. The archived v0.5.0 baseline is at DOI 10.5281/zenodo.23403300 (concept DOI 10.5281/zenodo.20614908 resolves to the latest version). This development manuscript includes subsequent corrections; its current evidence is identified by the source digest and verification ledger.

2 Introduction

Eye movements are a rich, non-invasive window onto attention, cognition, and oculomotor health. Three signal families dominate research use: the **gaze trajectory** (where the eyes point over time), **saccades** (the fast ballistic movements between fixations, with characteristic direction, amplitude, and peak-velocity dynamics), and **pupil diameter**. Primary experiments associate pupil diameter with memory load in a specific human task and with neuromodulatory activity in mouse cortex (Kahneman and Beatty 1966; Reimer et al. 2016). These associations motivate measurement; pupil diameter is not a specific readout of human cognitive load or a single transmitter. Dedicated infrared eye-trackers measure all three at high precision, but their cost restricts access; commodity cameras offer reach but force software to confront poorer optics, lower frame rates, browser/camera permission boundaries, illumination changes, and weaker calibration. A maturing open-source ecosystem now estimates related signals from these imperfect inputs: browser trackers such as WebGazer (Papoutsaki et al. 2016), large webcam and mobile gaze datasets such as GazeCapture (Krafka et al. 2016) and MPIIGaze (Zhang et al. 2017), appearance-based gaze estimators such as L2CS-Net (Abdelrahman et al. 2022), landmark-based capture stacks such as MediaPipe Iris (Vakunov et al. 2020), velocity- and dispersion-threshold event classifiers (Salvucci and Goldberg 2000; Nyström and Holmqvist 2010; Dar et al. 2021), microsaccade detectors (Engbert and Kliegl 2003), file-oriented processing and quality-reporting pipelines (Krakowczyk et al. 2023; Jakobi et al. 2024), and webcam or offline pupillometry tools (Shah et al. 2025; Zandi et al. 2021; Fuhl et al. 2015; Santini et al. 2018; Zhang and Jonides 2026; Mittner 2020; Cai et al. 2024).

Two engineering requirements motivate this implementation. First, capture through camera drivers, MediaPipe, browser WebRTC (Papoutsaki et al. 2016), or integrated platforms such as Pupil (Kassner et al. 2014) should have a boundary that allows analysis to run on ordinary arrays without hardware. Second, verification against constructed signals should be distinguished from agreement with independent measurements of real eyes. These are requirements for iTrace, not claims that existing packages lack modularity or validation. The established file-oriented tools already show the value of separating acquisition from analysis (Krakowczyk et al. 2023; Dar et al. 2021).

The existing tools each occupy a different point in this design space, and iTrace is positioned deliberately against them. WebGazer (Papoutsaki et al. 2016) pioneered in-browser webcam gaze but fuses capture, calibration, and estimation into one runtime. GazeCapture and MPIIGaze provide the public-data substrate for real-frame gaze evaluation (Krafka et al. 2016; Zhang et al. 2017), while L2CS-Net demonstrates how appearance models can predict pitch/yaw from unconstrained images (Abdelrahman et al. 2022). pymovements (Krakowczyk et al. 2023) offers a mature, file-oriented processing pipeline and now anchors explicit data-quality reporting standards (Jakobi et al. 2024), but it assumes the gaze samples already exist. REMoDNaV (Dar et al. 2021) is a strong event classifier whose four-class fixation / saccade / smooth-pursuit / post-saccadic-oscillation output is richer than ours, but it classifies an input signal rather than reasoning about how that signal was estimated from a camera. PupilSense/EyeDentify (Shah et al. 2025) addresses webcam pupil diameter with learned models and a dataset; PupilEXT, ElSe, and PuRe represent mature pupil-detection platform and algorithm lines (Zandi et al. 2021; Fuhl et al. 2015; Santini et al. 2018); PupEyes and pypillometry focus on reproducible pupil preprocessing and analysis after samples have been recorded (Zhang and Jonides 2026; Mittner 2020); Open-DPSM models luminance-driven pupil responses in dynamic stimuli (Cai et al. 2024). iTrace does not try to out-measure any of these on real eyes. It occupies a narrower gap: a *verification-first*, hardware-decoupled implementation of established gaze, saccade, microsaccade, pupil, and descriptive scanpath algorithms whose correctness is pinned to constructed ground truth and whose outputs can later be compared against public datasets, reference devices, or other packages when the caller supplies truth.

iTrace makes three contributions. First, it implements established algorithms in a pure NumPy/SciPy core with no hardware

dependency, then verifies every detector against synthetic signals whose events are known by construction. Second, it keeps webcam frames, MediaPipe iris landmarks, live HTML rendering, and dashboard/figure backends in optional shells whose outputs are ordinary typed gaze/pupil/capture records. The landmark-to-gaze mathematics is therefore testable without a camera, and the browser never becomes a second analysis implementation. Third, it adds bounded interfaces around that verified core: descriptive event statistics, maximum-likelihood distribution fitting, scanpath spread and entropy metrics, a bootstrap interval on the main-sequence exponent, a deterministic publication-figure gallery, a pure eye-crop pupil-segmentation helper that reports pixels or pupil/iris-relative units only, a benchmark report for caller-supplied truth files, and a guided live empirical workflow that writes derived records rather than raw eye video. The graphical summary in fig. 1 shows the same separation visually. sec. 3 describes the architecture and algorithms; sec. 14 reports the ground-truth verification, N=1 local empirical diagnostics, and figure gallery; sec. 22 and sec. 23 state the accuracy boundaries that still constrain webcam eye tracking.

3 Methods: architecture

iTrace is organised as two strata with a one-way dependency: a pure analysis **core** and an optional hardware **shell**. The split is not cosmetic — it is the mechanism that makes the scientific claims of this paper checkable.

The **core** depends only on NumPy, SciPy, and pandas. It divides into a signal-and-event layer (`geometry`, `calibration`, `velocity`, `saccades`, `mainsequence`, `pupil`, `pupilseg`, `pupilphase`, `encoding`, `eyemodel`, `scene`, `power`, `benchmark`, `experiments`, `pipeline`, `io`) and a quantitative `stats` subpackage layered on top of the detected events (`stats.descriptive`, `stats.distributions`, `stats.scanpath_metrics`, `stats.similarity`, `stats.timeseries`; sec. 11, sec. 12). Every value the core consumes or produces is a plain array or a typed dataclass; it has no knowledge of cameras, files beyond CSV, or rendering. The unit and coordinate conventions are fixed at the boundary — pixels with a top-left origin, degrees of visual angle with a mathematical-convention direction (`0deg` right, `+90deg` up), seconds for time, and a pupil unit that travels with every pupil number — so no ambiguous quantity ever leaks between layers.

Three recent additions keep that contract explicit at the places where a reader might otherwise overinterpret output. `pupilseg` accepts a supplied eye crop and returns a dark-component segmentation in image pixels, or a pupil/iris-relative sample if the caller provides an iris-radius normalizer; it does not infer millimetres and never claims absolute pupil diameter. `benchmark` accepts user-supplied truth and comparator event files and reports interval-overlap recovery, timing error, and amplitude error; every payload records the truth boundary instead of implying that detector agreement is a reference measurement. `experiments` adds the analogous file-replayable shape for prompted live sessions: it defines trial schedules, slices derived capture samples, and reports session quality plus held-out target residuals without storing raw video or treating screen prompts as a reference tracker. These modules are additive interfaces around the same core streams, not special cases that bypass the typed unit contract.

Calibration is kept in that same core layer because it is an analysis transform, not a camera driver. The shipped model is deliberately inspectable — a fitted two-dimensional affine map from raw gaze coordinates to known target coordinates, with RMS, percentile, and maximum error summaries. That makes screen-space calibration auditable when target points exist, while leaving live webcam output honestly labelled as relative/algorithmic when no external target or reference device is present.

The **shell** (`capture`, `live`, `dashboard`, `cli`, `viz`) is the only code that touches OpenCV, MediaPipe, Streamlit, Plotly, or matplotlib, and it imports those dependencies **lazily, inside functions**, never at module load; a missing optional dependency raises an actionable error rather than breaking `import itrace`. Two consequences follow, and they are the load-bearing design decisions of the whole package:

1. `import itrace` and the *entire* test suite succeed on a headless machine with none of the optional dependencies installed.
2. The scientific algorithms can therefore be verified continuously, in CI, without a webcam — which is precisely what makes their correctness auditable (sec. 18).

The `stats`, viz, and method layers obey a second, finer rule that keeps the headless guarantee honest: matplotlib is treated like any other optional plotting backend and lives **only** in the visualisation modules, imported at the top of those modules with the `Agg` backend selected before `pyplot`. A method or statistics module never imports a plotting library, so a fitted distribution, a scanpath-similarity matrix, or a windowed event-rate series is computed and returned as arrays regardless of whether any display is available.

The dependency rule is strict and one-directional: the shell may import the core, the core may never import the shell. A new capture backend (browser WebRTC, a deep-learning gaze model, a research IR camera) is added by writing a shell (sec. 8) that emits the same `GazeStream` / `PupilStream` / capture-record tables the core already analyses. The verified core never changes, every analysis in sec. 5 through sec. 12 applies unaltered to its output, and the visualization/report layer can be regenerated from the same stored payloads. The full module map and dependency contract are in sec. 27.

4 Methods: gaze geometry

The geometry layer turns raw image measurements into calibrated quantities in degrees of visual angle (dva). Pixel offsets from screen centre are converted by the exact relation

$$\theta = \arctan(s_{\mathrm{cm}}/d_{\mathrm{cm}}) \tag{1}$$

where s_{cm} is the offset in centimetres (pixels times the screen's cm-per-pixel) and d_{cm} the viewing distance. The inverse of eq. 1, `deg2pix`, is exact, so `pix2deg`$\circ$`deg2pix` round-trips to floating-point tolerance.

For appearance-based capture, normalised iris displacement within the eye aperture is mapped to eyeball rotation through a sphere model,

$$\theta_{\mathrm{gaze}} = \arcsin(o \cdot \sin \theta_{\max}) \tag{2}$$

with $o \in [-1, 1]$ the normalised iris offset and $\theta_{\max}$ the angle at full deflection; the map (eq. 2) is monotone and odd. Head-distance dependence is removed by dividing offsets by the inter-ocular distance in pixels and rescaling to a fixed reference, so the same physical movement yields the same value regardless of how far the subject sits from the camera.

Two conventions are fixed package-wide to prevent sign errors: gaze direction uses 0° = right and $+90^\circ$ = up (image-y, which grows downward, is negated on the way in); all non-finite inputs raise rather than silently coercing to zero, so a NaN can never masquerade as a valid centred gaze.

5 Methods: velocity and event detection

5.1 Velocity

Gaze velocity is estimated two ways, matching how data arrives. For uniformly sampled streams, the implementation uses a Savitzky–Golay local-polynomial derivative (Savitzky and Golay 1964) as a deterministic smoothing-and-differentiation choice; the window is clamped to an odd length not exceeding the trace. For non-uniform timestamps a central-difference gradient is taken on the actual time vector. Two-dimensional speed is the Euclidean norm of the per-axis velocities.

The default velocity dispatcher checks timestamp spacing before selecting its estimator: intervals constant within a relative tolerance of 0.001 use the uniform path; irregular intervals use the actual time vector. Differentiation rejects nonfinite, duplicate, and decreasing timestamps. Stream containers retain such clocks for quality diagnostics, but they cannot be treated as valid timing for event dynamics.

5.2 Fixations and saccades (I-VT, I-DT)

I-VT (Salvucci and Goldberg 2000) labels each sample saccadic when its 2-D speed exceeds a velocity threshold and collapses contiguous same-label runs into events; runs shorter than a minimum duration are reabsorbed into the surrounding fixation, suppressing single-sample velocity spikes. In noisy or intermittently tracked streams the configurable `merge_gap_s` parameter optionally bridges short subthreshold holes inside one high-velocity movement before the duration filter is applied. Only finite, analysable samples can be bridged; missing gaze and nonfinite velocity split the candidate intervals and are never labelled as fixations. The historical merge default remains zero. **I-DT** (Salvucci and Goldberg 2000) greedily extends a window while its spatial dispersion $(\max x - \min x) + (\max y - \min y)$ stays below threshold, emitting one fixation per maximal low-dispersion window. It likewise splits windows at nonfinite gaze samples.

5.3 Microsaccades (Engbert–Kliegl)

Microsaccades follow Engbert and Kliegl (2003). Velocity uses their five-sample moving-average estimator, and a per-axis elliptic threshold

$$(v_x/\eta_x)^2 + (v_y/\eta_y)^2 > 1 \tag{3}$$

(eq. 3) flags an event when held for at least a minimum number of samples. The threshold $\eta = \lambda\sigma$ uses the **median-based** robust scale estimator $\sigma^2 = \mathrm{med}(v^2) - \mathrm{med}(v)^2$, clamped at zero, so a few velocity outliers cannot inflate the threshold the way a plain standard deviation would — a property the test suite checks directly (sec. 14).

6 Methods: saccade dynamics and scanpath encoding

6.1 Main sequence

For each saccade iTrace reports amplitude, direction, duration, and peak velocity. The amplitude–peak-velocity relationship — the *main sequence* — is fit to two standard parameterisations: a saturating exponential

$$V = V_{\max}\,(1 - e^{-A/C}) \tag{4}$$

(eq. 4; Bahill et al. (1975)) by non-linear least squares, and a power law $V = aA^b$ by linear regression in log-log space, reporting the coefficient of determination in log-velocity space. The exponent depends on amplitude range, task, sampling, and model choice; iTrace uses it as a descriptive fit, not a clinical marker or a universal normal range. The nonlinear fit constrains $V_{\max}$ and C to

be positive and reports nonconvergence explicitly instead of substituting starting values. `fit_power_law` computes the log-log fit independently, so resampling that exponent does not require an unrelated saturating-model optimisation.

6.2 Scanpath encoding

Saccade sequences are encoded as direction characters as an implementation-level summary: `R/L/U/D` for the nearest cardinal, upper-case for long saccades and lower-case for short ones relative to a length threshold. N-gram statistics over the resulting string give a compact order-sensitive descriptor for package reports and tests without claiming biometric identification or anomaly-detection performance.

7 Methods: pupillometry

The pupil pipeline is a sequence of pure transforms over a `PupilStream`, each independently testable. It follows the conservative preprocessing vocabulary common to open pupil-analysis tools: blink handling, interpolation, baseline-correction, quality reporting, and transparent preprocessing records (Zhang and Jonides 2026; Mittner 2020). Blinks are detected as runs of NaN or sub-threshold samples and removed by linear interpolation with edge padding, using elapsed timestamps rather than sample indices, so the partial-occlusion ramp on either side of a closure does not leak into the signal; a trace with no valid samples is reported as unusable rather than silently flattened to a constant. The same rejection applies when padding removes every interpolation anchor. Outliers are flagged by the scaled median absolute deviation (1.4826 MAD), which is robust to the very spikes it must catch; when MAD is zero, finite deviations from the median are flagged explicitly. The deblinked trace is low-pass filtered with a deterministic zero-phase Butterworth implementation choice (with a short-trace moving-average fallback) (Butterworth 1930) and baseline-corrected, subtractively or divisively, against a chosen pre-event time window. The short-trace fallback uses edge padding and a bounded odd window so it preserves length and constant input. The actual second-order-section padding requirement determines whether Butterworth filtering is available. A cutoff at or above Nyquist also uses the moving-average fallback; that output is not represented as Butterworth-filtered.

The runtime `PupilConfig` now controls the same decisions the standalone functions expose: the low-confidence pupil threshold, interpolation padding, and Butterworth cutoff/order and the fallback window (`smooth_window`). Reports include both cleaned-signal summaries (mean/standard deviation/min/max, phase peak and trough counts) and raw-quality summaries (valid fraction, blink fraction/count, median sample interval, peak dilation velocity, and peak constriction velocity). The live webcam pupil channel therefore remains a relative proxy, but the analysis path records how much of the trace was usable and how strongly the cleaned pupil changed over time. Brightness and contrast effects are not modelled away in this version; tools such as Open-DPSM show why dynamic visual confounds need explicit modelling when pupil size is interpreted cognitively (Cai et al. 2024).

The separate `pupilseg` module covers a narrower, testable image-space step for callers that already have an eye crop. It thresholds the finite crop intensities, selects the largest dark connected component, reports centroid/radius/area and contrast-derived confidence, and converts that segmentation into either a `PupilUnit.PIXELS` sample or a pupil/iris-relative sample when an iris-radius normalizer is supplied. This deliberately does less than established pupil detectors and platforms such as ElSe, PuRe, or PupilEXT (Fuhl et al. 2015; Santini et al. 2018; Zandi et al. 2021): it is a pure-core fixture-verified path for pixels/relative measurements, not a millimetre calibration or device validation claim.

For closed-loop and online use, a causal `PhaseDetector` classifies each streamed sample as dilation, constriction, peak, or trough using only the current and prior samples (Kronemer et al. 2025). Causality is not merely asserted but verified: feeding the detector a prefix of a signal reproduces exactly the labels it assigns to that prefix in a full run, proving no future sample influences a past label (sec. 15).

This causality statement applies to `PhaseDetector` itself. Blink interpolation and zero-phase filtering use surrounding samples; the offline pupil pipeline and the rolling reanalysis displayed by the live dashboard are not causal filters.

8 Methods: the capture shell

The capture shell is the sole hardware-facing module, and it is deliberately thin. Its testable heart, `iris_landmarks_to_sample`, converts one frame of normalised MediaPipe Face Mesh landmarks — supplied as a plain float array — into a binocular gaze sample, with no MediaPipe import anywhere in the call. MediaPipe Iris is cited here as a commodity-camera landmark source, not as a guarantee of calibrated eye-tracker accuracy (Vakunov et al. 2020). Because the function consumes plain floats, the entire landmark→gaze mathematics is unit-tested headless, and the same function is the bridge the forward-model loop (sec. 9) exercises.

`WebcamSource` wraps the genuinely device-bound work: opening the camera with OpenCV, running MediaPipe inference per frame, and yielding timestamped capture samples with gaze, a relative pupil/iris proxy, frame index, an FPS estimate, and quality flags. The live-frame path preserves that existing sample API while also returning the frame dimensions, a clamped pixel-space eye-region box derived from the Face Mesh eye and iris landmarks, and a JPEG data URI containing the zoomed eye crop. The proxy is explicitly labelled relative: MediaPipe Face Mesh does not expose a calibrated pupil boundary or millimetre diameter, so the live path records a camera-dependent size signal rather than a physical pupil measure. `itrace record` writes the detected gaze samples to CSV, can write the relative pupil proxy to a second CSV, and can write the full capture-record table (`frame_index`, monotonic timestamp, gaze, pupil proxy, FPS estimate, and quality flags) via `--records-out`. `itrace camera-probe` exercises dependency and camera access without turning hardware availability into a scientific validation claim. Both hardware commands suppress noisy

native OpenCV/MediaPipe stderr diagnostics by default while retaining `--backend-logs` for debugging. Constructing it without the `capture` extra installed raises a clear, actionable error naming the missing extra, rather than a bare `ModuleNotFoundError` from deep in the stack. Only the few lines that actually grab frames are excluded from coverage; the dependency-absent error path, landmark conversion, capture-sample shape, split gaze/pupil CSV writers, and full capture-record CSV writer are all tested headless.

The additive binocular path keeps the existing public `iris_landmarks_to_sample` API intact while exposing `iris_landmarks_to_binocular` for callers that need per-eye diagnostics. It returns the binocular mean gaze plus right/left `EyeGazeDiagnostic` records, horizontal vergence, vertical disparity, asymmetry, and the averaged relative pupil proxy. These quantities are quality and confidence signals derived from the same landmark frame; they are not a binocular calibration or stereoscopic validation study.

The separate `itrace live-html` command adds a local single-user FastAPI orchestrator around the same verified Python path. The browser page receives WebSocket messages containing the eye crop, the latest capture sample, rolling `pipeline.analyze_session` summaries, event records, and plot-ready series; the browser itself only draws controls and native Canvas/SVG diagnostics. The eye crop is therefore a transient local-display payload, not a persisted artifact in the default workflow. It is configured as the `web` extra and remains import-safe: `fastapi`, `uvicorn`, OpenCV, and MediaPipe are imported only inside server or capture entry points. Persistence is explicit: without `--output-dir` the app is in-memory, and with `--output-dir` it writes CSV/JSON artifacts only when live export is requested or a guided empirical protocol completes. Calibration follows the same ownership rule. The HTML controls call start/sample/fit/reset endpoints, but `LiveState` owns the session points: for each displayed target the backend aggregates recent finite gaze samples, stores the median raw point with a sample count, and fits `AffineCalibration` only after enough target points exist. The browser draws the result; it never implements a competing calibration estimator.

The same backend-owned pattern now supports a guided empirical-session workflow. `itrace.experiments` defines a default derived-record protocol: fixed-center gaze, natural reading, and center/four-corner target prompting, each 30 s by default. The browser displays targets and countdowns, while the server records absolute trial windows in `LiveState`, slices the derived `CaptureSample` stream, and automatically writes per-trial gaze, pupil, and capture-record CSV files plus a protocol manifest and JSON report after the final required trial. The report estimates session-specific sampling regularity, finite-gaze fraction, pupil-valid fraction, fixed-gaze dispersion, drift slope, calibration residuals on held-out target intervals, and target acquisition latency. It deliberately persists derived records only: no raw eye video or eye-crop image sequence is written by the default workflow, and the target residuals are labelled as prompted-session diagnostics rather than reference-device accuracy.

9 Methods: 3-D forward model and the closed loop

Detectors verified against a one-dimensional position signal are only as auditable as the path that produced that signal. To exercise the geometry/landmark path end-to-end — the part a real webcam pipeline actually runs — we add a 3-D forward model (`eyemodel`) and an animated scene (`scene`) that drive the *real* estimator from synthetic anatomy with known truth.

9.1 The eyeball-to-landmark forward model

A parameterised eyeball — centre, radius, gaze yaw/pitch, and pupil and iris radii — is rotated to a true gaze direction. The iris ring and pupil disc are placed on the sphere surface at that orientation and **perspective-projected** through a pinhole camera to the normalised, MediaPipe-shaped landmark array the capture layer consumes. The anatomical eye corners are fixed: they sit on the face, not on the rotating globe, so they do not move with gaze. Because the corners are stationary while the iris translates across the visible sclera, the iris-to-corner offset encodes gaze exactly as a real face mesh would, which is what lets the same `geometry` routine that an actual recording exercises be the one under test. The projection is a forward map from a physical pose to image coordinates; it is deliberately not the inverse the estimator computes.

9.2 The animated scene

The scene (`scene`) drives a deterministic trajectory of fixation targets joined by minimum-jerk saccades — the same velocity profile the synthetic single-saccade generator uses — together with a pupil baseline carrying Gaussian dilation events and blink intervals. At each frame it emits both the per-frame 3-D truth (gaze yaw/pitch, pupil size, blink state) and the projected landmark array, so the entire recording has a ground truth that no detector could have peeked at.

9.3 The closed loop

The **closed loop** then feeds those landmarks back through the production estimator — `iris_landmarks_to_sample` reconstructs a `GazeStream`, which the standard `pipeline` analyses into fixations, saccades, and a pupil summary — and measures recovery against the 3-D truth: gaze angular error in degrees, saccade detection against the scripted saccades, and pupil correlation with the scripted baseline.

The design is deliberate on one point: the forward model (3-D sphere + perspective projection) and the estimator (the arcsine sphere approximation of sec. 4) are **independent formulations**, not two halves of one identity. Their agreement therefore tests the geometry rather than restating it — a tautology would yield exactly zero residual — while the small non-zero residual is the price of the estimator's approximation, an honest, bounded quantity rather than a hidden one. What this construction can and cannot establish is taken up in sec. 16 and sec. 23: it certifies that the analysis code recovers truth from a geometrically faithful

but *idealized* eye, and it is silent — by construction — about corneal refraction, the kappa angle, lens distortion, and every other physical effect the pinhole model omits.

10 Methods: noise model and statistical design

10.1 Idealized noise model

To ask *how much noise the recovery tolerates*, we identify the quantity that dominates corruption. Photon shot noise, JPEG compression, pixel quantisation, and MediaPipe's finite precision all manifest downstream as one thing: **jitter in the normalised image coordinates of the landmarks**. We model it as an *idealized* additive **i.i.d. Gaussian** of standard deviation σ (normalised image-coordinate units) on every landmark before estimation. This is explicitly simplified: it ignores spatial correlation between adjacent landmarks, temporal correlation, heteroscedasticity (error grows at eccentric gaze and under poor lighting), and systematic bias — and it is *not* derived from sensor optics. Systematic *biases* (corneal refraction, kappa angle, lens distortion) are excluded entirely; they belong to the device validation gap (sec. 23). Appearance-based gaze work on GazeCapture, MPIIGaze, and L2CS-Net makes the same practical point from the opposite direction: real-frame accuracy depends on dataset, camera, illumination, head pose, and calibration conditions, not just on the downstream event detector (Krafka et al. 2016; Zhang et al. 2017; Abdelrahman et al. 2022). Pupil-size measurement, which a real pipeline derives from a separate segmentation we do not implement, carries its own observation noise scaled with σ as a stated modelling assumption — so the pupil channel's noise behaviour is *assumed*, not emergent. The model is chosen to be the simplest perturbation that is falsifiable rather than the most realistic one. The recovery curves are conditional sensitivity results; omitted error sources do not make them a quantitative lower or upper bound on a real webcam's performance.

10.2 Statistical design

The sweep is a **sensitivity / robustness** analysis, not a hypothesis-testing power calculation. At each noise level we run n independent, seeded closed-loop recordings (sec. 9) and measure three outcomes — gaze RMS error (deg), saccade detection F1, and pupil correlation with truth. Per level we report the mean and a 95% **bootstrap percentile** confidence interval across the n trials. We use a bootstrap rather than a symmetric Student-t interval because two of the metrics are bounded (F1 $\in [0, 1]$, correlation $\in [-1, 1]$): near 1.0 a symmetric interval can run past the bound, whereas a percentile of resampled means cannot leave the range of the in-bound observations. We then locate the noise level at which each outcome crosses a usability bound by linear interpolation between grid points. The interval describes variability of the mean across the random landmark-noise realisations sampled by the n seeded trials (a Monte Carlo uncertainty interval over the noise process), not variability from re-running the deterministic pipeline. Saccade F1 scores a detected event as a true positive when its sample interval **overlaps** a ground-truth saccade interval (no separate temporal tolerance); precision and recall are computed from the same overlap match and combined as the harmonic mean, so a detector that fragments one true saccade into several events or merges several into one is penalised rather than rewarded. Every trial seed is deterministic, so the whole sweep — and therefore every figure and table number — is reproducible. Full statistical detail is in sec. 26. The sweep is falsifiable: were recovery independent of noise, gaze RMS would be flat across the grid.

Event-level resampling in sec. 11 has a different sampling unit from these whole synthetic trials. Its IID form assumes exchangeable detected events; ordered moving blocks preserve some local dependence under approximate stationarity, with an explicitly chosen block length (Künsch 1989). Neither form resamples participants. Support-respecting endpoints alone do not establish nominal coverage (Efron 1987). The design, failed-draw counts, and interpretive limits are stated in sec. 28.

For package-level validation, `itrace.validation` separates two cases that are often conflated. Synthetic-domain validation has event truth: seeded clean, webcam-jitter, head-drift, and low-light/dropout sessions provide saccade intervals, amplitudes, directions, pupil events, blinks, and quality flags, so within-domain repeat summaries and cross-domain stability gaps can report interval-overlap precision, recall, F1, amplitude error, direction error, peak-velocity error, and pupil-validity statistics. Live webcam diagnostics do not have that oracle. Following the data-quality reporting emphasis in recent eye-tracking standards work, the live HTML interface therefore reports sampling regularity, finite gaze fraction, pupil-valid fraction, path length, dispersion, and warnings, but deliberately leaves recovery F1 undefined unless a future reference-device or public-dataset path supplies truth (Jakobi et al. 2024).

11 Methods: descriptive and distribution statistics

The `itrace.stats` package turns the typed events of sec. 5 and sec. 6 into summary statistics, fitted distributions, and spread metrics. Like the rest of the core it is pure NumPy/SciPy, takes explicit seeds wherever it resamples, and operates on whatever gaze it is given — synthetic streams with known ground truth or recorded ones. The methods below *describe* a scanpath; none of them measure real-eye accuracy, which remains the device validation gap of sec. 23. The generated interpretation ledger in fig. 11 is a companion artifact for these methods: it maps each reported statistic to its source payload, estimand, scholarship basis, and explicit non-claim boundary.

11.1 Descriptive summaries

Given a list of fixations or saccades, `stats.descriptive` returns the count and the mean, standard deviation, median, and interquartile range of the relevant per-event quantity — fixation duration for fixations; amplitude, duration, direction, and peak velocity for saccades — reusing the property arrays already vectorised by `saccades.saccade_properties`. An empty list yields a count of

zero and zero-filled summaries rather than raising, so a recording in which a detector finds no events of a given type flows through the pipeline without a special case. These are ordinary sample statistics over the detected events, not estimates of an underlying population the recording was not designed to sample.

The publication diagnostics sidecar adds a robust amplitude-shape block before any fitted model is interpreted. It reports the median, inter-quartile range, coefficient of variation, mean-minus-median gap, Bowley's quartile skewness, and Tukey-style 1.5 IQR outside-value counts (Bowley 1901; Tukey 1977). The same block adds seeded percentile-bootstrap intervals for the median and IQR (Efron and Tibshirani 1993), reported as finite-sample uncertainty summaries rather than population confidence claims. These descriptors are intentionally redundant with the parametric fits below: the quartile summaries show the scale and asymmetry of the detected events even when the sample is too small, too skewed, or too structured for a fitted family to deserve much interpretive weight.

11.2 Distribution fitting and model comparison

The publication payload compares five marginal candidates for detected positive amplitudes: **normal, gamma, log-normal, Weibull, and ex-Gaussian**, fitted by SciPy maximum-likelihood routines. Gamma, log-normal, and Weibull have positive support with location **fixed at zero** (`floc=0`), leaving shape and scale free. Normal and ex-Gaussian have support on the whole real line and can assign mass to impossible negative amplitudes; they are unconstrained descriptive comparators, not positive-truncated models. The ex-Gaussian is a Gaussian convolved with an exponential, parameterised by SciPy as $(K, \mathrm{loc}, \mathrm{scale})$ with $K = \tau/\sigma$ (Ratcliff 1979; Luce 1986). The free-parameter count excludes fixed location parameters, including a caller-fixed normal or ex-Gaussian location. Models are ranked by the **Akaike and Bayesian information criteria** ($\mathrm{AIC} = 2k - 2\ln\hat{L}$, $\mathrm{BIC} = k\ln n - 2\ln\hat{L}$) (Akaike 1974; Schwarz 1978), which penalise the extra ex-Gaussian parameter against any gain in fit, and a lower-is-better ordering is reported alongside the raw log-likelihoods. The diagnostics payload also reports the small-sample correction $\mathrm{AICc} = \mathrm{AIC} + 2k(k+1)/(n-k-1)$ when it is defined, plus normalised Akaike weights $w_i \propto \exp(-\Delta_i/2)$ and evidence ratios within the same finite candidate set (Burnham and Anderson 2002). These weights are relative support diagnostics for the compared families; they are not posterior probabilities and do not prove that the selected family generated the data. Here n counts detected events, not gaze frames. AIC remains the declared ranking, weighting, and winner-frequency criterion; AICc and BIC are sensitivity descriptors. AICc is undefined for $n \le k + 1$, and this commonly used correction is a heuristic for these fitted marginal families, not a finite-sample guarantee for every density model (Hurvich and Tsai 1989). The marginal product likelihood also does not model serial dependence between events. The same sidecar runs a deterministic bootstrap over detected amplitudes, refits the same candidate family set on each resample, and records the lowest-AIC winner frequencies as a stability diagnostic, not as posterior model probabilities (Efron and Tibshirani 1993; Burnham and Anderson 2002). Absolute goodness of fit is summarised by a one-sample **Kolmogorov–Smirnov** statistic $D = \sup_x |F_n(x) - \hat{F}(x)|$ between the empirical CDF and each fitted CDF (Massey 1951); we report D as a descriptive distance and treat the companion fixed-CDF p-value as **uncalibrated**, since the parameters were estimated from the same sample. A separate refit-bootstrap KS diagnostic simulates from the fitted family, refits each draw, and uses the plus-one Monte Carlo tail probability. Its family was selected from these same data: this exploratory diagnostic is not adjusted for model selection, serial dependence, or multiple comparisons (sec. 28). Fitting requires a minimum number of finite positive observations and raises a `ValueError` below it, mirroring the guard in `mainsequence.fit`.

For the publication diagnostics payload, the lowest-AIC family is also rendered as a quantile–quantile check: empirical order statistics are plotted against the fitted distribution's quantiles at plotting positions $(i - 0.5)/n$ (Wilk and Gnanadesikan 1968). The sidecar reports the QQ residuals, root-mean-square residual, central-80% residual, maximum absolute residual, fitted-line slope, intercept, and correlation. The same best-fit family is also checked on the CDF scale: empirical plotting positions are compared with fitted CDF values at the observed amplitudes, producing a P-P-style residual series, residual RMSE, mean absolute residual, maximum residual, and the fitted KS distance (Massey 1951). The same residual block reports a Cramer–von-Mises-style integrated squared CDF distance and an Anderson–Darling-style tail-weighted distance (Anderson and Darling 1952), plus lower- and upper-tail maximum residuals. These scalar distances help separate a fit that is wrong everywhere from one that fails mainly in the tails, but they are still descriptive because the family and parameters were selected from the same events. For scale, the payload also reports a DKW/Massart reference width `epsilon = sqrt(log(2 / alpha) / (2 n))` for the empirical CDF at $\alpha = 0.05$ (Massart 1990), plus whether the maximum fitted-CDF residual exceeds that width. This band is displayed as a distribution-free visual reference for ECDF fluctuation for IID observations around a fixed CDF; it does not account for serial dependence. Because the distribution family and its parameters were selected from the same sample, it is not a formal acceptance test. The quantile and CDF residuals are model-adequacy diagnostics for the detected sample; they are not a proof that the family is true or that another recording would share the same amplitude distribution.

11.3 Scanpath spread metrics

The spatial extent of a scanpath is summarised by four metrics computed over fixation centroids (in degrees of visual angle). **Gaze dispersion** is the root mean square distance of fixations from their centroid. The **bivariate contour ellipse area (BCEA)** — the area of the ellipse expected to contain a stated fraction (default 68%) of fixations under a bivariate-normal assumption, `BCEA = 2 * pi * k * sigma_x * sigma_y * sqrt(1 - rho^2)` with $k = -\ln(1 - P)$ — is a standard, units-of-deg^2 measure of fixation stability introduced for fixational eye movements (Steinman 1965) and widely used as a low-vision fixation-stability index (Castet and Crossland 2012). The **convex-hull area** of the fixation centroids gives a distribution-free footprint of explored space via `scipy.spatial.ConvexHull` (returning zero for fewer than three non-collinear points rather than raising). Both BCEA and convex-hull area are reported only as conditional descriptors: BCEA assumes approximate bivariate normality, which a structured scanpath violates, and is flagged as such.

Two **entropy** metrics capture organisation rather than extent. *Stationary position entropy* discretises fixation centroids onto a coarse spatial grid and reports the Shannon entropy of the occupancy histogram in bits — higher when gaze is spread evenly across cells, lower when it concentrates. *Direction-transition entropy* reuses the cardinal scanpath string of sec. 6: it forms the first-order transition-count matrix between successive direction symbols and reports the entropy of the transition distribution, a compact index of how predictable the scan order is, in the spirit of information-theoretic scanpath analysis (Boccignone and Ferraro 2004). Both are reported in bits, without maximum-entropy normalisation. Direction-transition entropy is the count-weighted conditional entropy $H(\text{next} \mid \text{current})$, not joint transition entropy. Spatial binning defaults to the supplied record's extent; cross-record comparisons must specify a common grid, extent, and symbol convention. Unequal event counts and sparse transition rows introduce estimation uncertainty and finite-sample bias (Paninski 2003). Zero observed transitions do not establish zero population probability. Both metrics return zero for degenerate one-cell or one-symbol inputs.

11.4 Bootstrap CI on the main-sequence exponent

The power-law main-sequence fit of sec. 6 reports a point estimate of the exponent b (`power_b` from `mainsequence.fit_power_law`). To attach uncertainty, `stats.scanpath_metrics.main_sequence_exponent_ci` resamples the matched amplitude / peak-velocity pairs with replacement B times (default $B = 2000$, explicit seed), refits the log-log regression on each resample, and reports a two-sided 95% **bootstrap percentile** confidence interval on b (Efron and Tibshirani 1993). We use the percentile interval here for the same reason it is used in the noise sweep (sec. 26): it makes no symmetry assumption and stays inside the range of the resampled estimates. Coverage is approximate and not established merely by that property. IID resampling perturbs the observed event pairs under an exchangeability assumption; moving-block resampling is available with an explicit block length. Neither estimates a population of eyes from a single recording. The detailed result reports attempted, successful, and failed draws. Rank-deficient draws are excluded and the surviving interval is conditional on successful fits; if no valid draw survives, the interval is unavailable and the API raises instead of returning zero-width uncertainty (Morris et al. 2019).

12 Methods: advanced detection and scanpath comparison

The detectors of sec. 5 fix one threshold for a whole recording and reduce a scanpath to summary scalars. Two further layers relax those choices: an `itrace.detection` module that adapts its thresholds to the data and reports movement dynamics the fixed detectors discard, and two `itrace.stats` modules that compare and resolve scanpaths over time. Like the rest of the core they are pure NumPy/SciPy, take explicit seeds wherever they resample, and operate on whatever gaze they are given — synthetic streams with known ground truth (sec. 9) or recorded ones. Nothing here measures real-eye accuracy; that remains the device validation gap of sec. 23.

12.1 Adaptive detection and movement dynamics

The fixed-threshold I-VT of sec. 5 requires the analyst to pick a velocity cut that suits the recording's noise floor. `detection.detect_ivt_adaptive` instead follows the data-driven scheme of Nyström & Holmqvist (Nyström and Holmqvist 2010): it seeds a peak-velocity threshold, iteratively re-estimates it from the mean and standard deviation of the samples that fall *below* the current threshold (the putative fixation/noise population), and stops when the estimate converges. The converged value, not a hand-set constant, then classifies saccades, so the same call adapts to a clean high-rate trace and a noisy webcam one without re-tuning. The converged threshold is also surfaced in `SessionReport.quality` when the configurable pipeline is run with `DetectionConfig(method="adaptive_ivt")`, so a downstream report can state which threshold actually certified the detected events.

The same configurable route also exposes a conservative merge-gap repair. When `merge_gap_s` is positive, I-VT bridges only subthreshold gaps whose duration is no longer than that value and whose run is bounded on both sides by saccadic samples. This is deliberately narrower than smoothing the whole velocity trace: it repairs one known failure mode — short capture dropouts or plateaus splitting one saccade — while preserving event boundaries elsewhere and making the chosen repair window visible in the report configuration.

Fast eye movements are followed by **post-saccadic oscillations** (PSOs) — the brief wobble of the eye as it settles — which a plain I-VT either swallows into the saccade or mislabels as a tiny second saccade. `detection.detect_pso` scans the samples immediately after each detected saccade offset for a short, lower-amplitude velocity excursion in the settling window and tags it as a PSO rather than a movement, in the spirit of the glissade/PSO class that REMoDNaV adds to the saccade/fixation dichotomy (Dar et al. 2021). iTrace does not yet claim REMoDNaV's full four-class smooth-pursuit classification — that is named as future work in sec. 23 — so the PSO tag is reported as a labelled sub-event of the saccade it follows, not as an independent event class.

Two further quantities describe the *dynamics* the fixed detectors leave on the floor. `detection.intersaccadic_intervals` returns the gaps between successive saccade onsets (seconds), the inter-saccadic-interval distribution used throughout the eye-movement literature to characterise scanning tempo and fixational dwell (Holmqvist et al. 2011); an empty or single-saccade input yields an empty array rather than raising. `detection.saccade_peak_accelerations` differentiates the velocity signal a second time and reports the peak acceleration (deg/s^2) within each saccade interval, the second-order companion to the peak-velocity main sequence of sec. 6. Both reuse the velocity estimators of sec. 5 rather than recomputing them, so a recording's acceleration profile is consistent with the velocities its saccades were detected from.

12.2 Scanpath similarity

Comparing two scanpaths — a recording against a template, or two sessions of the same task — needs a distance, not a scalar summary. `itrace.stats.similarity` provides three complementary ones over the cardinal direction strings of sec. 6, demonstrated on recovered scanpaths in sec. 21.

The **Levenshtein scanpath distance** is the minimum number of single-character insertions, deletions, and substitutions that turn one encoded direction string into another, computed by the standard dynamic-programming edit-distance recurrence over the two strings (Levenshtein 1966); it captures how much one scan *order* must be rewritten to match the other and is robust to small local insertions a fixed alignment would punish. We report both the raw edit count and a length-normalised version (distance divided by the longer string's length) so paths of unequal length are comparable, and an empty-versus-empty comparison returns distance zero.

The **n-gram cosine similarity** lifts the comparison from order to habitual sub-patterns. Each scanpath is turned into a vector of n-gram counts over its direction string — reusing `encoding.ngram_counts` — and the two vectors are compared by cosine similarity over their shared vocabulary, giving a value in $[0, 1]$ that is high when two recordings share the same characteristic short sequences regardless of where they occur, following the vector-space comparison logic used in information retrieval (Salton et al. 1975). Either vector being empty (a recording with fewer than n saccades) yields a defined similarity of zero rather than a division by zero.

The **transition matrix** is the raw material both similarities are built to summarise: the first-order count (or row-normalised probability) of moving from each direction symbol to each next one, the same construction that underlies the direction-transition entropy of sec. 11 and the generative, information-theoretic view of scanpaths (Boccignone and Ferraro 2004). Exposing the matrix directly lets an analyst inspect *which* transitions drive a similarity or entropy value rather than trusting the scalar, and the row-normalised form is guaranteed to be a proper stochastic matrix or an all-zero row for an unobserved symbol.

12.3 Windowed event-rate time series

A single recording is rarely stationary: scanning tempo, fixation stability, and blink rate drift over a task. `itrace.stats.timeseries` slices a recording into fixed-width time windows (with an optional overlap) and reports, per window, the fixation rate, the saccade rate, and the blink rate in events per second, each built from the typed events the core already detected and the blink intervals of the pupil channel. The result is an aligned set of arrays — window-centre times and the three rate series — suitable for plotting a session's evolution or for feeding the spatial-stability metrics of sec. 11 (gaze dispersion, BCEA, used as a low-vision fixation-stability index (Castet and Crossland 2012)) a window at a time. Windows that contain no samples report a rate of zero rather than `nan`, and a window narrower than the sampling interval is rejected with an actionable error, so a mis-specified window size fails loudly instead of producing an empty or misleading series.

13 Graphical abstract

The package is intentionally composed as a narrow capture shell feeding a pure analysis core, with every live display downstream of Python-computed payloads rather than browser-side analysis. The graphical abstract (fig. 1) uses an eye-to-code metaphor: an observed eye is converted into typed records, the pure Python core performs the tested signal processing, and generated reports/export artifacts carry the evidence forward. Its badges intentionally name only the current gate, the five-session diagnostic v1 scope, and the absence of any reference-device claim. The footer also records the MIT license, fixed GitHub release URL, and the Zenodo DOI for this version. The figure communicates architecture and evidence boundaries; it is not a validation figure for webcam accuracy.

14 Results: ground-truth recovery

The first question for any detector is whether it recovers events it could not have peeked at. Every trace in this section is synthetic: a generator builds the signal *and* returns the parameters used to build it, so the recovered numbers are checked against truth held out by construction. None of these results speak to real-eye accuracy, which remains the device validation gap of sec. 23. They do, however, expose the kinds of implementation errors that matter before hardware is involved: time-unit mistakes, coordinate sign flips, threshold off-by-one boundaries, and accidental smoothing choices that move event onsets.

On a synthetic fixation→saccade→fixation trace at 250 Hz, the I-VT detector (Salvucci and Goldberg 2000) recovers exactly one saccade bracketed by two fixations; the recovered amplitude matches the constructed 10° to within 5% and peak velocity to within 10% across the test grid. The min-jerk generator sweeps amplitude and direction, and the recovered direction tracks the constructed angle around the full circle (the gaze convention of 0° right, +90° up), so a sign flip or axis swap on either path would surface immediately. A purely-noisy fixation trace yields zero saccades at the default threshold — the anti-criterion confirming the detector does not hallucinate events from noise.

The Engbert–Kliegl detector (Engbert and Kliegl 2003) recovers an embedded 0.5° microsaccade from fixational jitter while ignoring the surrounding noise; a probe confirms the threshold uses the median-based robust estimator (it returns a value below half the plain standard deviation in the presence of velocity outliers), the property checked directly in sec. 5. Recovered microsaccade amplitude and peak velocity fall on a tight, low-amplitude main-sequence cluster, distinct from the large-saccade branch — the expected separation of the two scales rather than a fitted claim. The main-sequence fit recovers the saturating $V_{\max}$ and C of a synthetic relationship (Bahill et al. 1975) to within 10% and returns a power-law exponent inside the physiological 0.4–0.9 range (fig. 2).

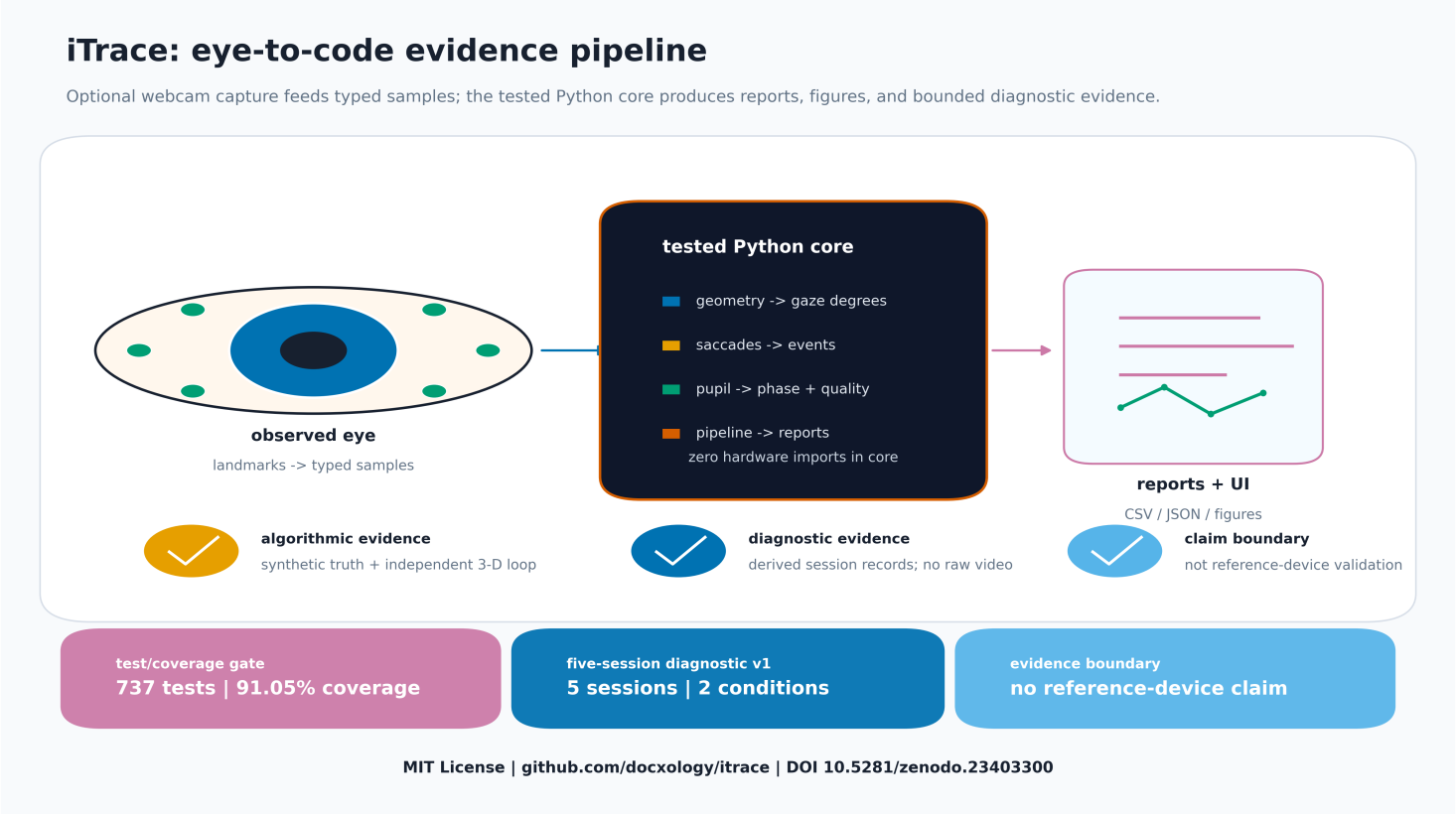


Figure 1: Graphical abstract of the iTrace eye-to-code evidence pipeline. A stylized observed eye feeds typed gaze, saccade, pupil, and quality records into the pure Python core, which then produces reports, figures, and live displays from Python-computed payloads. Three badges summarise the current verification gate, the five-session diagnostic v1 release scope, and the no-reference-device-claim boundary. The footer states MIT License, github.com/docxology/itrace, and the Zenodo DOI 10.5281/zenodo.23403300; the figure describes architecture and evidence boundaries, not webcam accuracy validation.

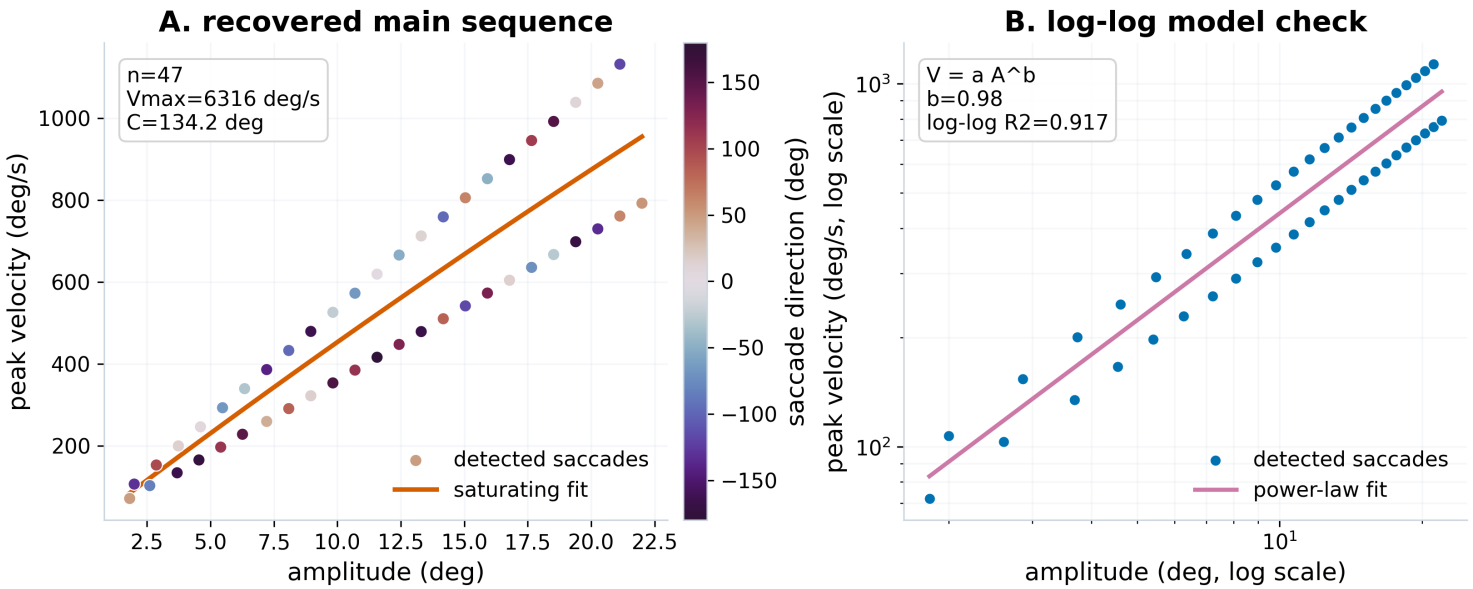


Figure 2: Main-sequence recovery from a synthetic multi-amplitude recording. Panel A plots detected saccades by amplitude, peak velocity, and direction, with the fitted saturating curve and recovered parameters. Panel B repeats the same detections on log-log axes with the fitted power law, making the exponent and goodness-of-fit explicit. The evidence is recovery of a seeded synthetic relationship and detector property extraction; it is not a physiological population estimate or real-device validation.

15 Results: pupillometry

A sinusoidal pupil signal with an injected NaN blink is correctly deblinked (no residual NaNs), and the interpolated-plus-smoothed trace coincides with the underlying sine away from the blink window — the blink is bridged, not fabricated. A fully-invalid trace is reported as unusable rather than silently flattened, so a recording the camera never resolved cannot masquerade as a flat pupil. These are recovery checks against a known generator, not statements about pupil-size accuracy on a real eye (sec. 23). That distinction matters because the pure pupil pipeline can verify blink handling, interpolation, phase causality, and unit propagation even when the live webcam pupil proxy remains a relative image-derived signal.

The causal phase detector labels peaks within a two-sample window of the analytic sine maxima, and the online-equals-offline-prefix test confirms it consults no future sample: replaying the signal one sample at a time reproduces the offline labels exactly up to each prefix, so the causality property of sec. 7 holds empirically rather than by assertion. The detected peak and trough counts match the number of sine cycles in the trace, the coarsest sanity bound on the detector. Saccade direction distributions over the same synthetic sessions are summarised as polar histograms (fig. 3).

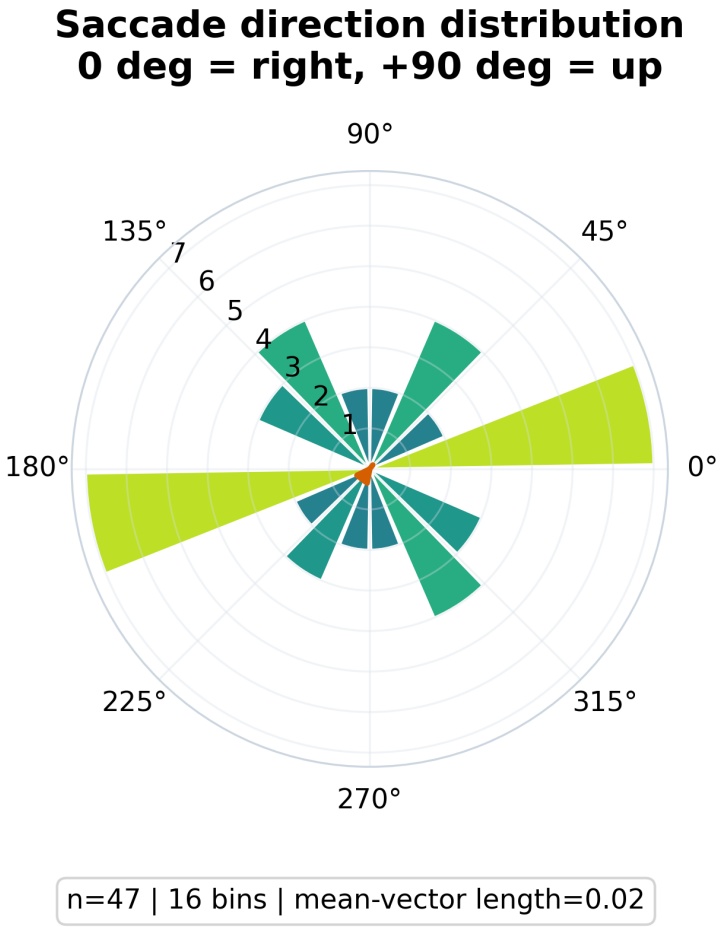


Figure 3: Saccade direction distribution over the synthetic multi-saccade recording. The polar convention is printed in the figure (0° = right, +90° = up), bars encode counts over 16 angular bins, and the orange resultant vector summarises directional balance. The panel verifies the package-wide direction convention and recovered event directions on synthetic data; it does not infer natural viewing bias.

16 Results: closed-loop recovery

Running the full loop — 3-D eyeball → perspective projection → landmark extraction → estimation → recovery — on the default animated scene (four fixations joined by three saccades, two pupil dilation events, one blink, at 120 Hz) recovers gaze with an RMS residual of **0.16**° against the 3-D truth (maximum 0.23°) for gaze within ±15°, and recovers the three scripted saccades exactly (fig. 4). The residual is small but strictly non-zero. Independence of the formulations is established by their separate equations and implementations (sec. 9); a nonzero residual alone would not establish independence.

This is an *internal-consistency verification*: the tested configurations can expose sign flips, axis swaps, unit errors, and coordinate-frame mismatches. It is **not** device validation or a proof for untested configurations. The 0.16° residual is conditional on the simulated scene. Shared idealisations (pinhole projection, no corneal refraction, a spherical eye, zero kappa) are absent from its error budget, but omitting them does not establish a mathematical lower or upper bound on real-device error (sec. 23). The recovered

pupil ($r = 1.0$) is a near-identity consistency check (the projected pupil/iris ratio is a monotone function of pupil radius), confirming the deblink/pipeline preserve the signal in this example, not pupil-size accuracy or agreement in millimetres.

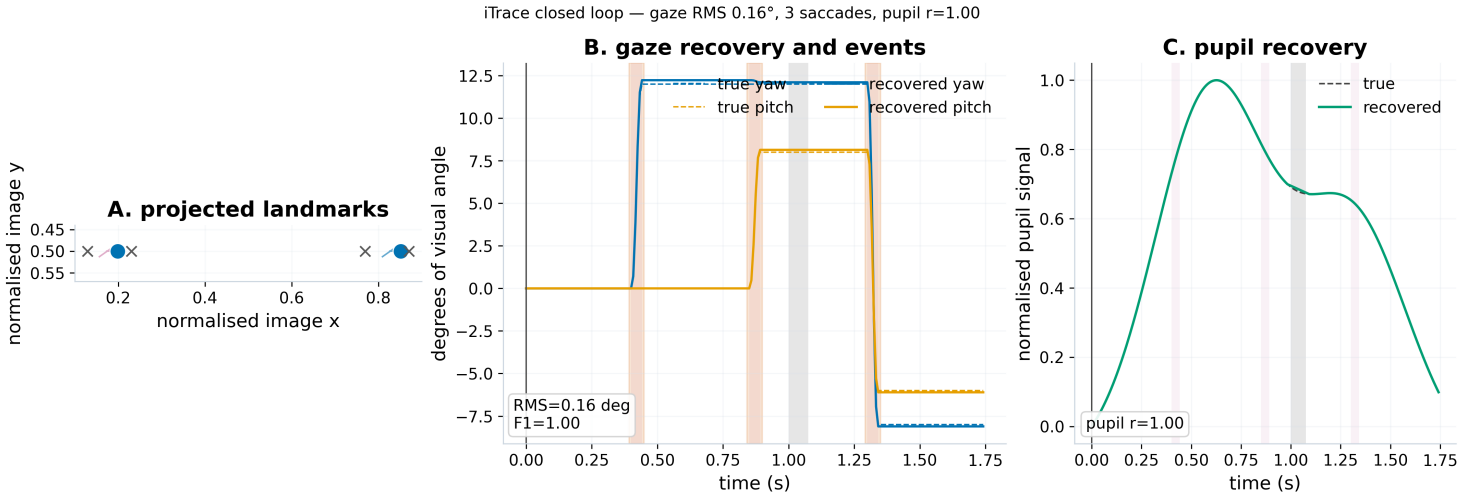


Figure 4: Closed-loop recovery on the default synthetic scene. Panel A shows projected binocular landmark geometry and iris-centre paths. Panel B overlays true and recovered yaw/pitch with scripted and detected saccade intervals. Panel C overlays normalised pupil dynamics and the blink interval. The independently formulated generator and estimator are checked on this scene; the residual is conditional on shared idealisations and is not a bound on measured webcam error.

The synthetic generator supplies all three target signals jointly — gaze direction, saccade intervals, and pupil diameter — animated as floating 3-D eyeball orbs (fig. 5).

17 Results: noise-sensitivity analysis

Sweeping the landmark-noise standard deviation σ from 0 to 0.016 (normalised image units), 25 seeded trials per level, gives a clear and differentiated degradation of the three signals (fig. 6, tbl. 1). The sweep is intentionally idealised: it injects independent landmark noise into the same synthetic scene rather than replaying real MediaPipe failures, so its value is to rank algorithmic fragilities under controlled perturbation.

Gaze RMS error rises monotonically and approximately linearly with σ, crossing the 2° usability bound at $\sigma \approx 0.0048$. **Saccade** detection is the most fragile emergent signal: F1 falls below 0.8 already at $\sigma \approx 0.0014$ under the revised one-to-one overlap scorer. Detection rests on velocity, the time-derivative of position, which amplifies high-frequency noise. Both orderings *emerge* from propagating landmark noise through the real estimator.

The **pupil** result carries a sharp caveat: its robustness (correlation above 0.9 until $\sigma \approx 0.005$) is **robust by construction** under the assumed `pupil_noise_scale`, not a measured property, and would move if that parameter changed. It is reported as a conditional illustration, not a finding; the genuine emergent result is the gaze-vs-saccade ordering.

The most defensible practical takeaway comes from translating σ to pixels. At a 640 px width, the saccade crossing is approximately 0.92 px, whereas the gaze crossing is approximately 3 px in this idealised sweep. That pixel-scale comparison is not a universal detector specification: real webcam landmark error is camera-, pose-, lighting-, compression-, and algorithm-dependent. It characterises overlap-based event recovery under this perturbation model. It does not establish onset/offset accuracy or a real-camera noise tolerance; those require independent measurements and boundary-error metrics.

The N=1 empirical target residual in sec. 19 is much larger than the residuals in this idealised landmark-noise sweep for the expected reason: it bundles calibration, screen-prompt, user-compliance, target-eccentricity, timestamp, and webcam-session effects into one prompted-target diagnostic. That makes it useful for bounding realistic live-demo and stress-test ranges, but it should not be back-solved into an iid landmark-noise σ or treated as reference-device gaze accuracy.

The same numbers, with per-cell uncertainty, are the generated statistics table (tbl. 1); it and the figure are generated from one sweep so they never drift. (Snapshot at $n = 25$; regenerate with `scripts/generate_power_figure.py`.)

Table 1: Recovery vs landmark noise sigma: mean [lower, upper] pointwise interval.

sigma (norm)	sigma (px@640)	gaze RMS (deg)	saccade F1	pupil r
0.0000	0.00	0.163 [0.163, 0.163]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
0.0010	0.64	0.452 [0.445, 0.457]	0.949 [0.920, 0.977]	0.996 [0.996, 0.996]
0.0020	1.28	0.842 [0.830, 0.855]	0.614 [0.586, 0.640]	0.982 [0.982, 0.983]
0.0040	2.56	1.665 [1.640, 1.692]	0.222 [0.215, 0.228]	0.934 [0.931, 0.937]

sigma (norm)	sigma (px@640)	gaze RMS (deg)	saccade F1	pupil r
0.0080	5.12	3.418 [3.372, 3.464]	0.276 [0.259, 0.293]	0.800 [0.792, 0.808]
0.0160	10.24	6.798 [6.694, 6.898]	0.558 [0.514, 0.606]	0.546 [0.528, 0.565]

Recovery vs landmark noise sigma (n=25 seeded trials/level; mean [pointwise 95% percentile interval], approximate coverage). sigma in normalised units and pixels at 640 px width.

18 Results: cross-implementation agreement and gates

On a shared synthetic trace the core I-VT and microsaccade detectors agree with an independent reference implementation (`reference_impl`, raw finite-difference velocity and an explicit-loop microsaccade detector) on event count and onset timing, despite using different velocity formulations — a guard against a bug shared by a single implementation. The two paths share only their input and the published algorithm definitions, so agreement is evidence the implementation matches the algorithm, not that two copies of one mistake agree.

The complete toolchain is green and reproducible: 737 no-mocks tests pass with 91.05% statement-and-branch coverage, `ruff` and `ruff format` report no issues, and `mypy --strict` finds no type errors across the source tree. The test count, coverage, version, gate date, and render evidence are stored once in `docs/verification_metrics.json`; the manuscript renderer hydrates these tokens from that JSON so stale literal metrics are caught by tests rather than silently carried into public artifacts. The historical release ledgers are preserved separately at `docs/releases/v0.4.1-verification.json` and `docs/releases/v0.5.0-verification.json`; dated audit evidence is not rewritten to impersonate a new execution. `scripts/verify.py` records individual exit codes, combined logs, runtime prefixes, a source digest, JUnit results, and machine-readable coverage in a new evidence directory, stopping at the first failed gate. CI exercises the development gates and a separate core-only test matrix, and checks an installed wheel outside the source import path.

Synthetic-domain validation selects gaze preprocessing from finite-input masks before analysis. It fills bounded gaps of at most 0.08 s, reports actual sample interpolation and removal, and rejects longer bounded gaps. This benchmark policy is explicit and does not depend on whether a particular numerical-library version raises during analysis; the raw-recording pipeline retains missing data.

Every synthetic generator and Monte-Carlo trial is explicitly seeded with `numpy.random.default_rng`. Within the recorded runtime and locked dependencies, figures and table numbers can be reproduced from a clean checkout; re-running the figure pipeline twice produces identical PNGs, and the noise sweep of sec. 17 reproduces its table cells exactly. No test mocks a computation: detectors run on real arrays with known ground truth, file I/O uses real temporary files, and figures are asserted to write non-empty PNGs. The full module map, dependency contract, and test strategy are in sec. 27.

The documentation and visualization layer now follows the same evidence rule. `docs/TRACEABILITY_MATRIX.json` maps each public validation claim and manuscript figure to the documentation surface where it appears, the tested Python method or script that computes it, the backing test file, the generated evidence artifact, and the truth boundary. That matrix is deliberately machine-readable: tests verify its paths and pure-core symbols, so a future figure, README badge, or live-analysis claim cannot become orphaned prose detached from the tested methods underneath it.

The validation-domain suite extends that reproducibility contract from a single trace to stress domains. `itrace synthetic-validation` writes JSON for repeated clean, webcam-jitter, head-drift, and low-light/dropout sessions, summarising within-domain recovery and across-domain stability. The live HTML page calls the same Python route when the user runs synthetic validation from the browser; the browser only renders the returned statistics and does not reimplement event detection or recovery scoring.

Finally, `itrace benchmark` provides a release-ready shape for external validation without pretending such data are bundled. It loads caller-supplied truth and comparator event CSV files, can score a gaze CSV through the iTrace pipeline, and writes one JSON report with interval-overlap recovery plus timing and amplitude errors for each system. That design matters because it separates the *mechanism* for validation from the *evidence* required to make a validation claim: the package can compute the same metrics against a reference tracker, public dataset, or comparator detector, but the report must carry the supplied truth source and boundary. A high score against another detector is detector agreement; a high score against an independent reference device would be a different and stronger claim.

The live experiment report follows the same reproducibility pattern for real webcam sessions. `itrace live-html` auto-saves a derived experiment bundle after the final required guided trial, and `itrace experiment-report` rebuilds the empirical summary from the exported manifest plus capture-record CSV. That command path is intentionally file-based: the reported jitter, drift, finite fraction, calibration residual, and target latency can be regenerated without a browser or camera, while the payload still states that prompted screen targets are not a reference-device validation study. The manuscript-facing single-pilot summary in fig. 7 is hydrated from that report when it exists; when it does not, the same render path exposes pending or unavailable fields rather than replacing missing empirical data with guessed values.

19 Results: single-pilot empirical diagnostics

The live empirical workflow is included here as an **N=1, order-of-magnitude case study**, not as a participant study or device validation claim. One local webcam session records fixed-gaze, reading, and center/four-corner target trials; the backend stores

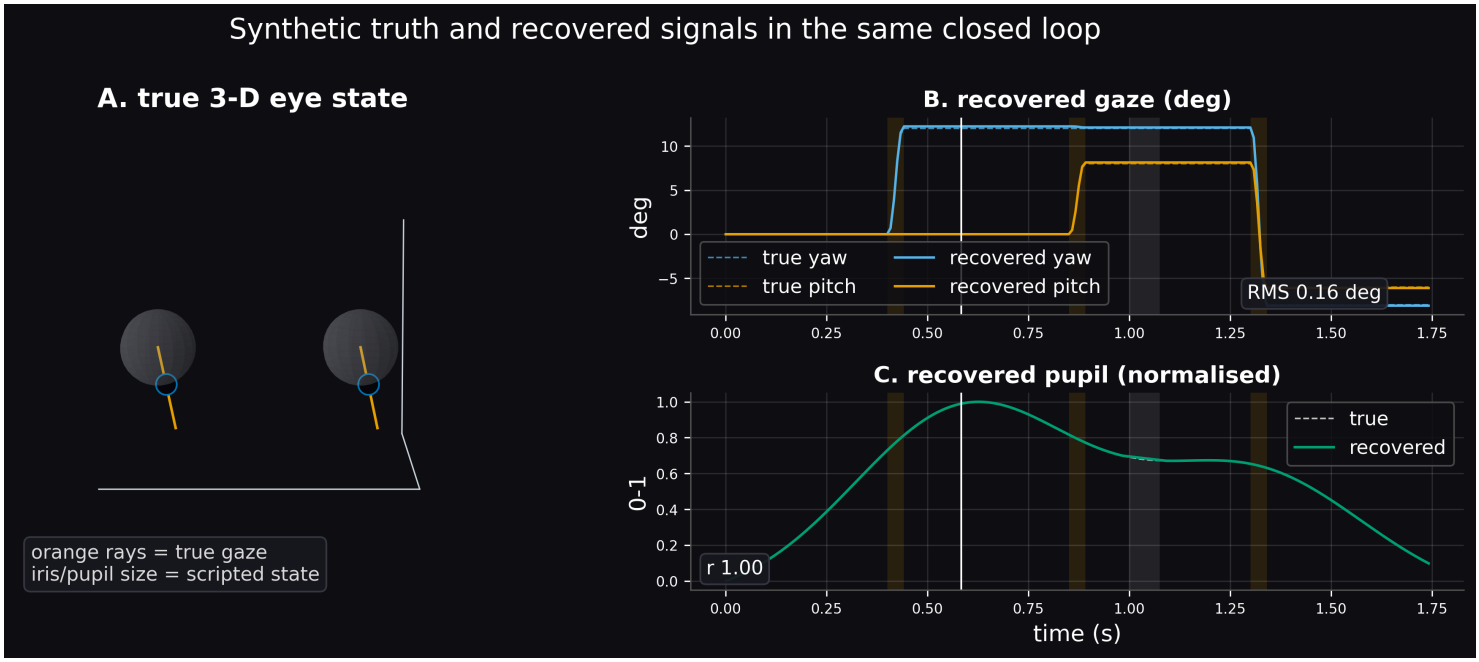


Figure 5: Synthetic truth and recovered signals rendered together. The 3-D panel shows the true eyeball orientation, gaze rays, iris, and pupil state at one representative frame; the trace panels repeat the recovered yaw, pitch, and pupil signals with true values, saccades, and the blink interval visible for audit. This panel is a visual audit of the synthetic scene and recovery path, not evidence from a recorded participant.

derived capture records, a protocol manifest, a target schedule, and an experiment report; and the manuscript renderer reads only the small summary JSON produced from that report. The current pilot source reports **3 completed trials, 2690 derived samples**. From that derived bundle, the renderer hydrates the session metrics as follows: finite-gaze fraction **100.0%**, sampling rate **29.7 Hz**, sampling-interval coefficient of variation **0.161**, maximum drift **0.054 deg/s**, held-out target RMS **6.79 deg**, and target acquisition latency **0.72 s median**. The values are stored in `docs/empirical_pilot_metrics.json`; when the pilot export is absent, the same render path leaves each field explicitly unavailable rather than filling a guessed value.

The development refresh retains the recorded empirical summary ledgers. The local derived session exports are not bundled in this checkout, so regenerating figures from those ledgers reproduces the recorded summary rather than a new empirical acquisition or reanalysis of the original records.

For adding more data, the release path now uses `docs/empirical_sessions_manifest.json` as an intake ledger. Each available session must name the de-identified participant, device, session group, replicate identifier, condition, protocol, consent scope, reference-evidence type, and repo-relative derived report path; planned sessions can be listed before their exports exist. A non-`none` reference-evidence type counts only when the row also names a valid repo-relative `reference_artifact` JSON; for the manual-annotation lane, that artifact stores sampled target-window judgments over derived report/record files rather than raw video.

The v1 empirical release is intentionally scoped as a single-participant, single-device, five-session diagnostic pilot, not as a device-validation study. The generated `docs/empirical_sessions_summary.json` and fig. 8 therefore treat participant/device counts as scope metadata while using five available sessions, five replicate IDs, and the two collected lighting conditions as the diagnostic readiness criteria. The current ledger reports **5 available sessions, 5 replicate IDs, 2 conditions, 0 reference-backed rows** and is **ready for diagnostic v1**. Under those checked-in diagnostic criteria, the minimum additional collection needed for v1 is **0** additional session exports, with **0** additional current-v1 condition(s) required.

The stronger 12-session, three-condition, reference-backed validation plan is preserved as future scope rather than counted as a current v1 blocker: **7 future session exports, 1 future condition (missing: indoor_office_backlit), and validated reference evidence still required**. Prompt-only replicates improve within-participant/device operating-scale coverage, but device validation still requires reference-device, public-dataset, or manual-annotation evidence backed by a valid artifact.

The pilot adds a different kind of evidence from the synthetic results. Synthetic truth recovery and the 3-D closed loop test the estimator under known gaze, events, and geometry (sec. 14; sec. 16); the noise sweep then shows how idealised landmark jitter propagates through the real estimator (sec. 17). The single local run instead contextualizes the scale of one real session: finite capture, webcam sampling regularity, short-session drift, prompted-target residuals, and UI/capture target acquisition latency. In that interpretation, the finite-gaze fraction and sampling CV are capture-stability checks, the drift value is a short-session stability check, the held-out RMS is a prompted-target residual scale, and the latency is a coarse task/UI timing quantity. None of these should be read as physiological saccade latency or reference-device gaze accuracy.

The pilot therefore sanity-checks one local operating scale for synthetic data generation without turning those variables into measured population parameters. The comparison is deliberately coarse, and fig. 9 renders the same comparison from a generated JSON sidecar

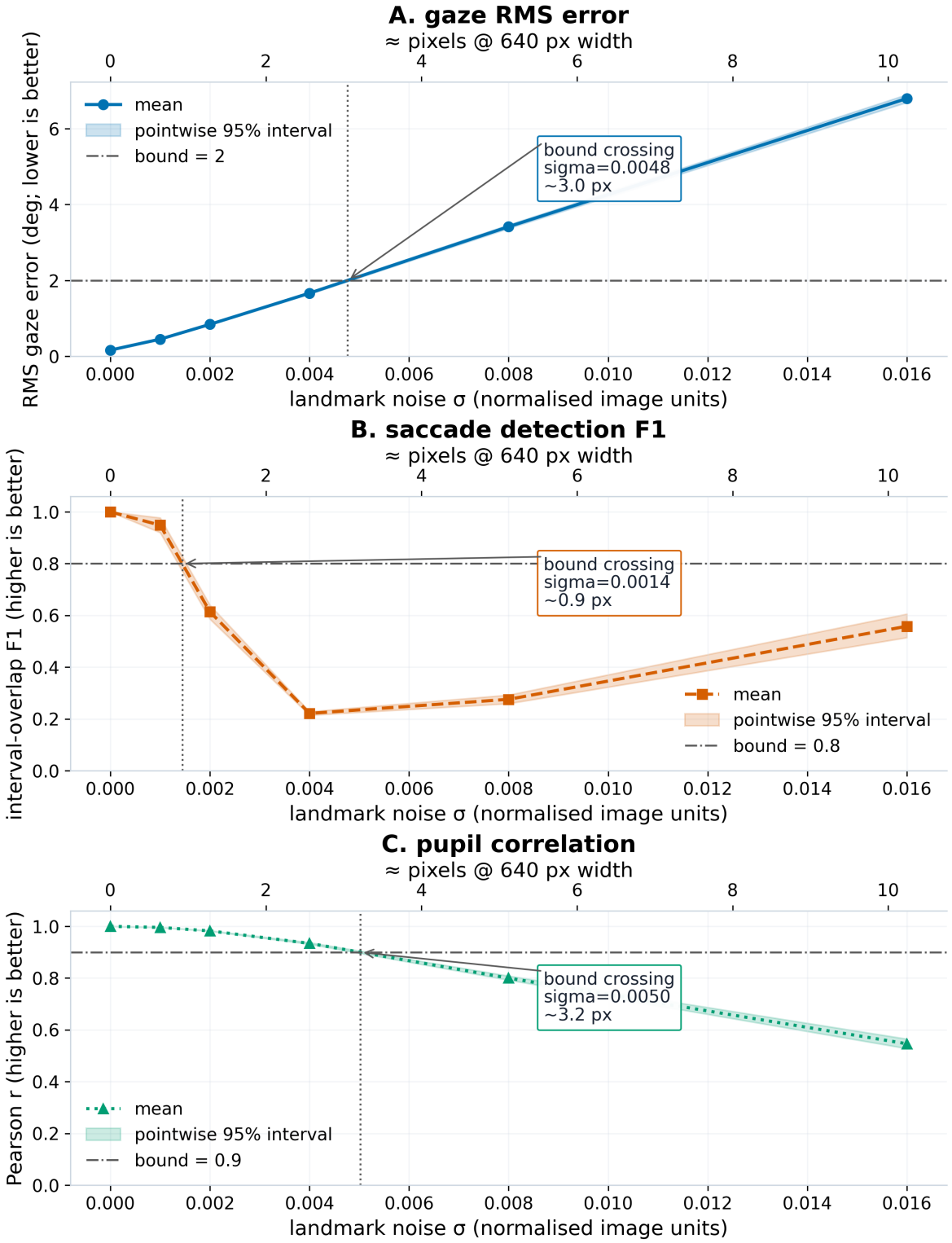


Figure 6: Recovery vs idealised webcam observation noise. Each panel reports the Monte Carlo mean and pointwise approximate 95% percentile interval for one recovered signal. These are not simultaneous confidence bands. The manuscript criterion and interpolated crossing are shown in normalised sigma and approximate pixels at 640 px width. Colour, markers, and line styles distinguish signals. Saccade F1 uses greedy one-to-one positive overlap; it does not measure onset/offset accuracy. Pupil correlation depends on the separate pupil-noise assumption.

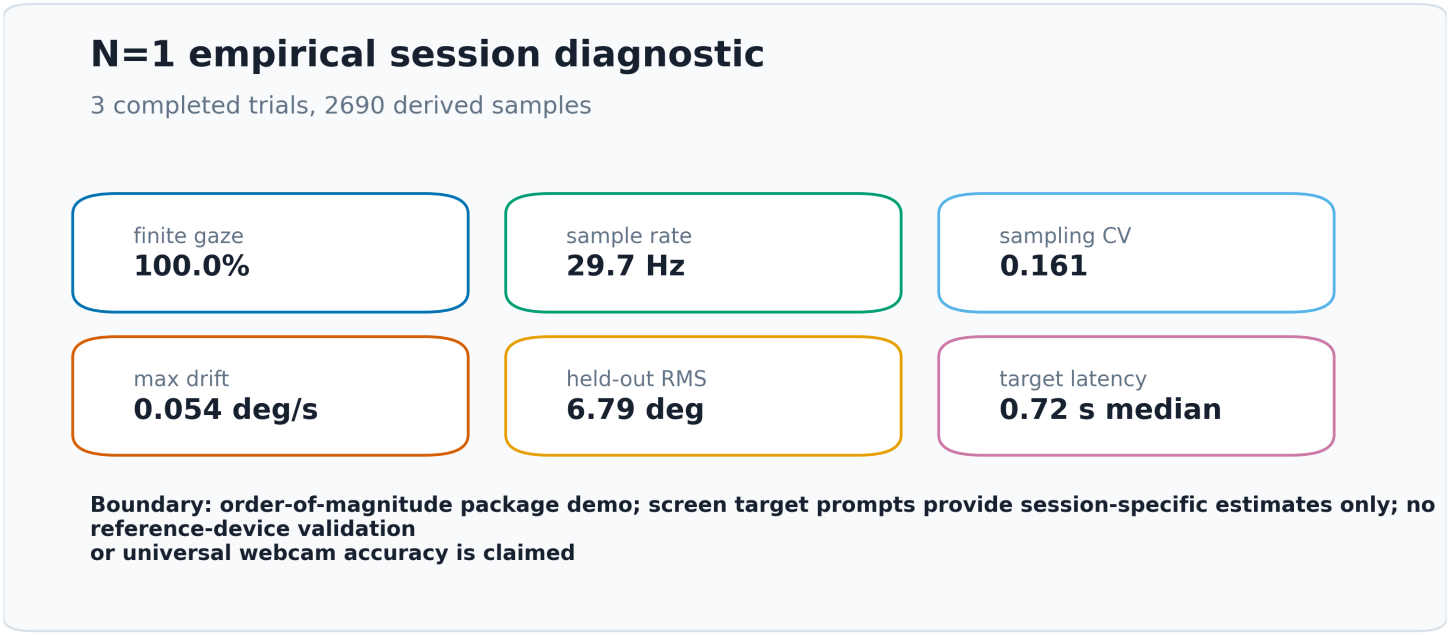


Figure 7: N=1 empirical session summary generated from the derived experiment report. The cards report finite-gaze fraction, sampling rate, sampling regularity, drift, held-out target residuals, and target acquisition latency as order-of-magnitude session diagnostics when a local pilot export exists; pending fields remain visible before recording. The figure demonstrates the package workflow and current-session quality ledger, not reference-device validation.

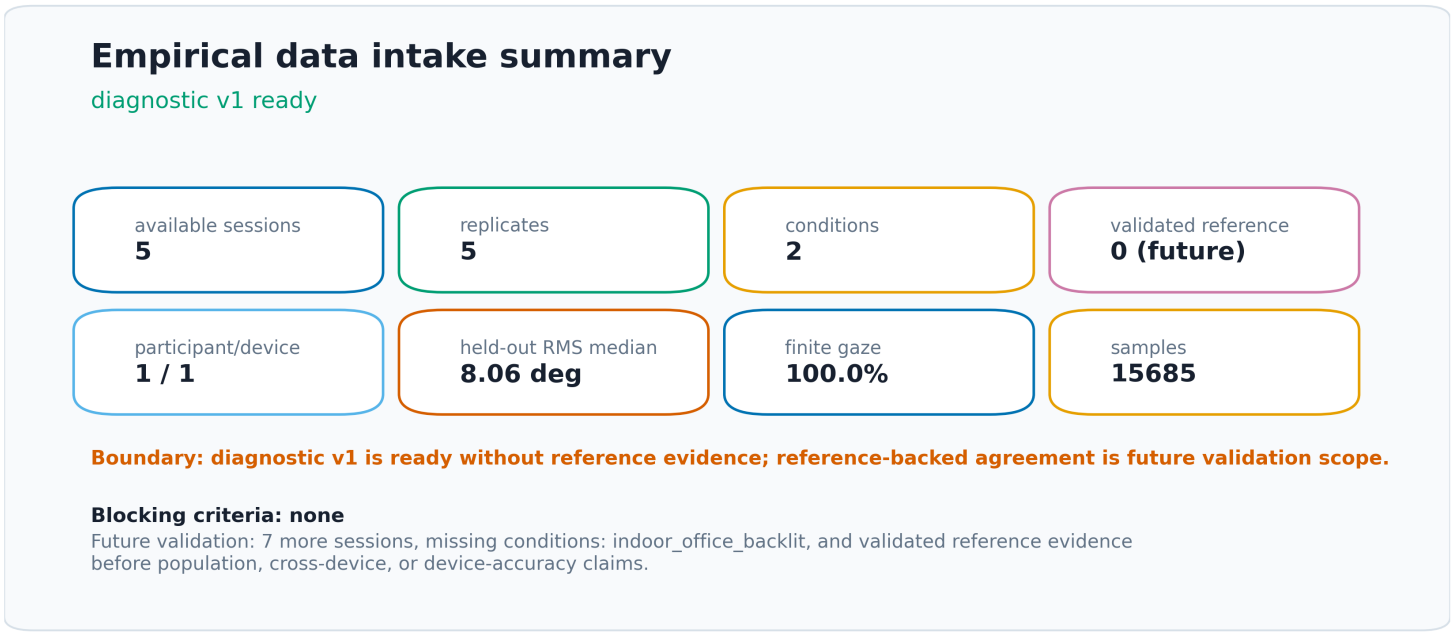


Figure 8: Empirical-session intake summary generated from `docs/empirical_sessions_manifest.json`. The figure reports available sessions, distinct replicates, conditions, reference-evidence count, participant/device scope, held-out residual scale, finite-gaze fraction, total derived samples, and current diagnostic-v1 status. The five-session prompt-only ledger is sufficient for diagnostic v1 readiness while population, cross-device, and device-accuracy claims remain future reference-backed validation scope.

so the visual and prose share one evidence boundary:

Table 2: N=1 pilot scale compared with synthetic/model variables.

Pilot diagnostic	Hydrated value	Context	Range interpretation
Finite gaze	100.0%	clean-session finite-sample examples; dropout domains	Supports the clean synthetic demos as plausible for a cooperative local run, while preserving low-light/dropout stress cases.
Sampling rate	29.7 Hz	live-style sampling cadence and timestamp grid	Contextualizes webcam-rate synthetic/session examples near a consumer-camera cadence rather than the higher-rate ideal traces used for algorithmic recovery.
Sampling CV	0.161	timestamp jitter in synthetic sessions	Gives an order-of-magnitude jitter target for live-style stress tests, separate from deterministic oracle traces.
Maximum drift	0.054 deg/s	head-drift and fixation-stability domains	Supports a low-drift baseline example; larger drift domains remain stress tests, not observed claims from this pilot.
Held-out target RMS	6.79 deg	prompted-target residual / calibration-bias scale	Contextualizes screen-prompt and calibration residual scale; it is not iid landmark noise, closed-loop residual, or gaze accuracy against a reference device.
Target acquisition latency	0.72 s median	target-schedule and UI/capture timing	Contextualizes task-response timing for the live protocol; it is not physiological saccade latency.

This comparison is intentionally asymmetric. The synthetic closed loop reports a near-ideal residual because both truth and measurements are generated inside a controlled model, while the N=1 pilot folds in calibration quality, user compliance, webcam timing, landmark behaviour, screen geometry, and target eccentricity. A much larger prompted-target RMS in fig. 9 is therefore not a contradiction of the synthetic validation; it is the practical scale at which the package's live workflow operates before any reference tracker is introduced. That makes the pilot useful for choosing realistic demo defaults, stress-test ranges, and caveats, while leaving real-device validation to a future public-dataset or reference-device study.

20 Results: figure gallery

The analysis core is paired with a deterministic visualization layer (`itrace.viz`, matplotlib on the headless Agg backend) that renders the standard eye-movement figures directly from a `SessionReport`. Each figure is produced from a synthetic session whose ground truth is known by construction, so the gallery is reproducible byte-for-byte and never depends on a captured recording. Categorical series draw from the shared Wong (2011) colour-blind-safe palette; scalar fields use perceptual sequential or diverging colormaps that match the quantity being encoded. All panels share the same house style — a readable font floor, white background, restrained spines, and compact result callouts — so the published figures match without pretending one palette serves every data type. In the multi-series comparison figures, signals are separated by marker and line style as well as colour, so the figures stay legible in greyscale. The figures below are regenerated for publication by `scripts/generate_figures.py`, which also writes the graphical abstract, noise sidecars, empirical summary, and figure manifest. The CLI command `itrace figures --out-dir output/figures --animations` remains the gallery-only refresh path.

The gallery spans the three signal families iTrace recovers. For **gaze and events**, a velocity trace (`output/figures/velocity_trace.png`) overlays the 2-D speed signal with the I-VT threshold and shades each detected saccade, while a spatial scanpath (`output/figures/scanpath.png`) draws fixations sized by dwell time and joined by fixation-transition arrows in screen coordinates. For **saccade dynamics**, an amplitude histogram (`output/figures/amplitude_histogram.png`) carries its

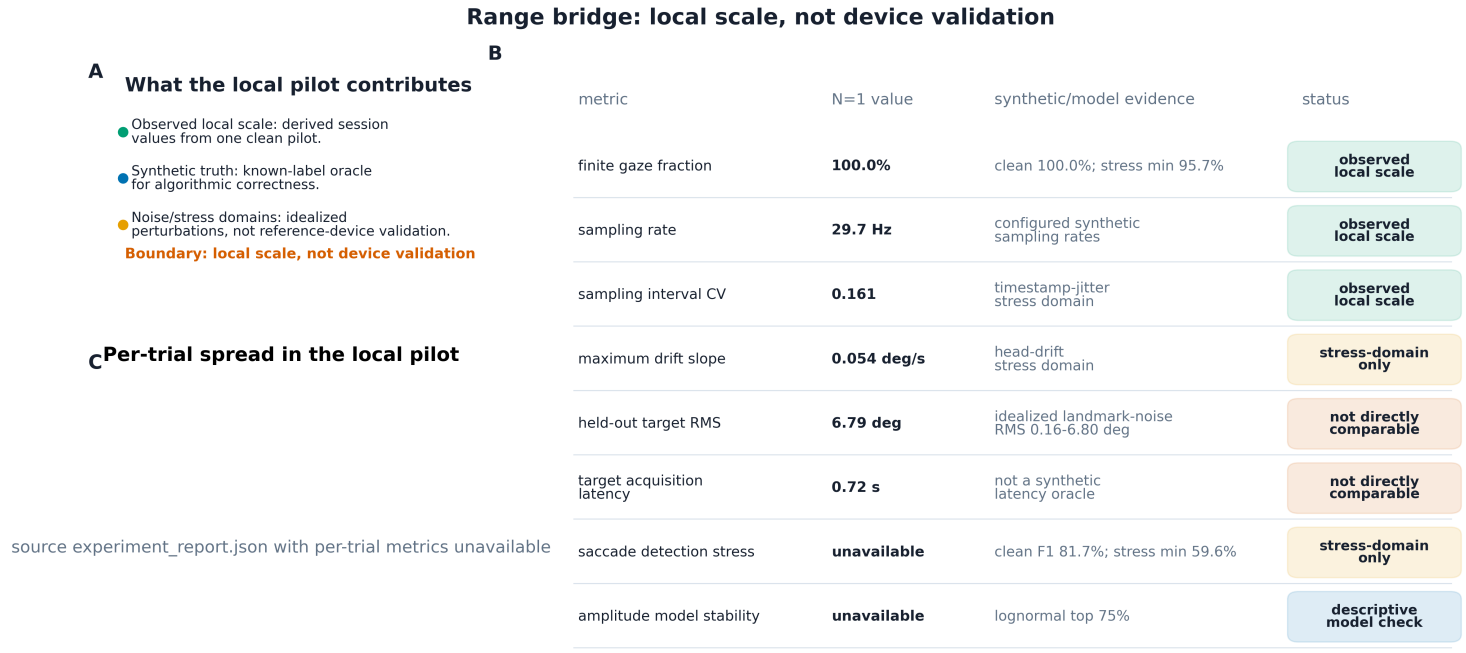


Figure 9: Synthetic-to-empirical range bridge generated from `output/figures/synthetic_empirical_range_bridge.json`. The rows place N=1 empirical session values beside synthetic-domain summaries, idealized landmark-noise sweep values, or descriptive statistical diagnostics, and each row is labelled as observed local scale, stress-domain only, or not directly comparable. The figure uses the pilot to contextualize synthetic defaults and stress ranges; it does not validate device accuracy, device performance, or webcam generality.

maximum-likelihood distribution overlay, and a main-sequence panel (`output/figures/main_sequence_diagnostics.png`) plots amplitude against peak velocity on log-log axes with the fitted power law and its residuals. For **pupillometry**, a pupil trace (`output/figures/pupil_trace.png`) shows the raw signal, the blink-interpolated and smoothed overlay, and the causally-detected dilation peaks and troughs.

A multi-panel session dashboard (`output/figures/session_dashboard.png`) composes the velocity trace, scanpath, amplitude distribution, main sequence, saccade-direction polar histogram, and a textual summary into a single at-a-glance figure. The statistical diagnostics composite in fig. 10 takes the same `SessionReport` view and makes the statistics layer inspectable: distribution-family ranking is plotted as lower-is-better AIC deltas with AIC weights, AICc/KS annotations, and robust amplitude-shape fields (median and IQR with seeded bootstrap intervals, Bowley skewness, and IQR outside values), plus a seeded bootstrap of the lowest-AIC family winner to show bootstrap lowest-AIC winner stability without treating that frequency as a posterior probability. The best-fit amplitude family is checked against empirical quantiles and fitted-CDF residuals with integrated/tail-weighted distance summaries and a DKW/Massart reference band, the main-sequence exponent is shown with exact seeded percentile endpoints and the successful-draw count, fixation centroids are paired with dispersion/BCEA/hull descriptors, and the scanpath string is decomposed into a first-order transition matrix. The panel is useful because each plotted value is computed by the tested Python statistics modules; it does not turn a synthetic session into population physiology or device validation evidence. The same refresh writes `output/figures/statistical_diagnostics.json`, so the PNG is display-only over an audited statistical payload rather than an untracked matplotlib-only calculation.

The statistical panel is intentionally information-dense, so fig. 11 records the companion interpretation ledger generated from `output/figures/statistical_interpretation_ledger.json`. Each row names the statistic, its reported value, the estimand it actually summarizes, and the claim it explicitly does not support. This matters for the scholarship boundary: AIC weights remain relative candidate-family weights, DKW bands remain finite-sample empirical-CDF reference bands, bootstrap intervals remain report-level uncertainty summaries, and the N=1 range bridge remains a single-session scale check rather than device validation.

The gallery now also includes gaze density, fixation heatmap, AOI dwell, event raster, pupil PSD, saccade-rate, microsaccade, and deterministic synthetic replay outputs. Because every panel is a pure function of a `SessionReport`, the same plots apply unchanged to a real recording analysed through the pipeline, subject to the same detector and calibration limits stated in sec. 23.

21 Results: scanpath comparison and temporal dynamics

The descriptive scanpath metrics of sec. 11 and the spatial/timeline visualisations of sec. 20 turn a single recording into a comparable summary of *where* and *when* gaze moved. As everywhere in this section, the inputs are synthetic sessions with known ground truth:

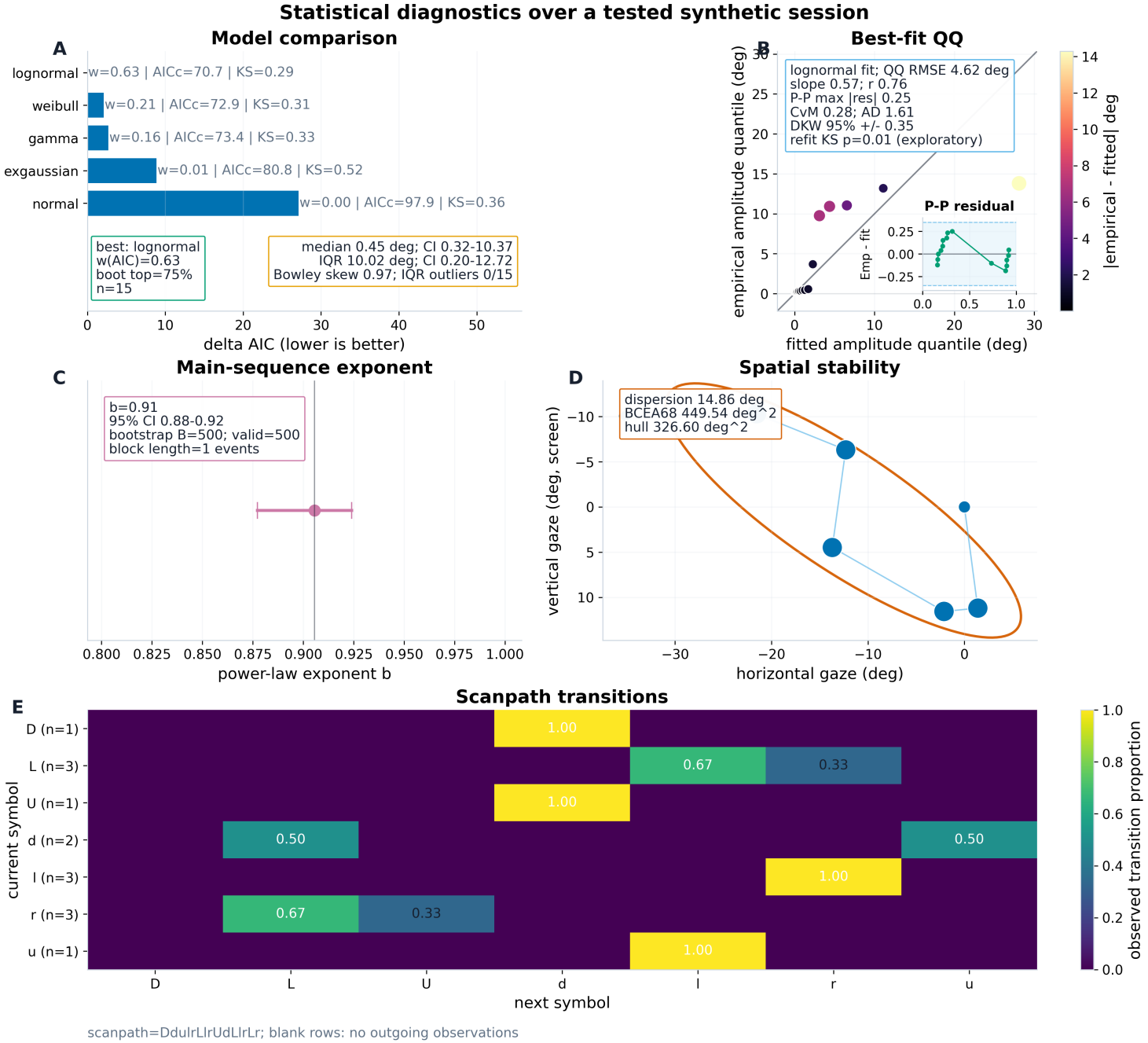


Figure 10: Statistical diagnostics over a deterministic synthetic session. Panel A ranks amplitude families by delta AIC with AIC weights, AICc/KS annotations, and event-shape summaries. Panel B shows QQ and fitted-CDF residuals, an IID DKW reference width, and an exploratory refit-bootstrap KS probability that is not adjusted for family selection. Panel C displays actual percentile endpoints for the main-sequence exponent and the successful resample count. Panel D shows fixation-centroid spread. Panel E displays empirical transition proportions with row counts; blank rows contain no observed outgoing transitions. These are conditional sample diagnostics, not real-eye accuracy, population inference, or model acceptance.

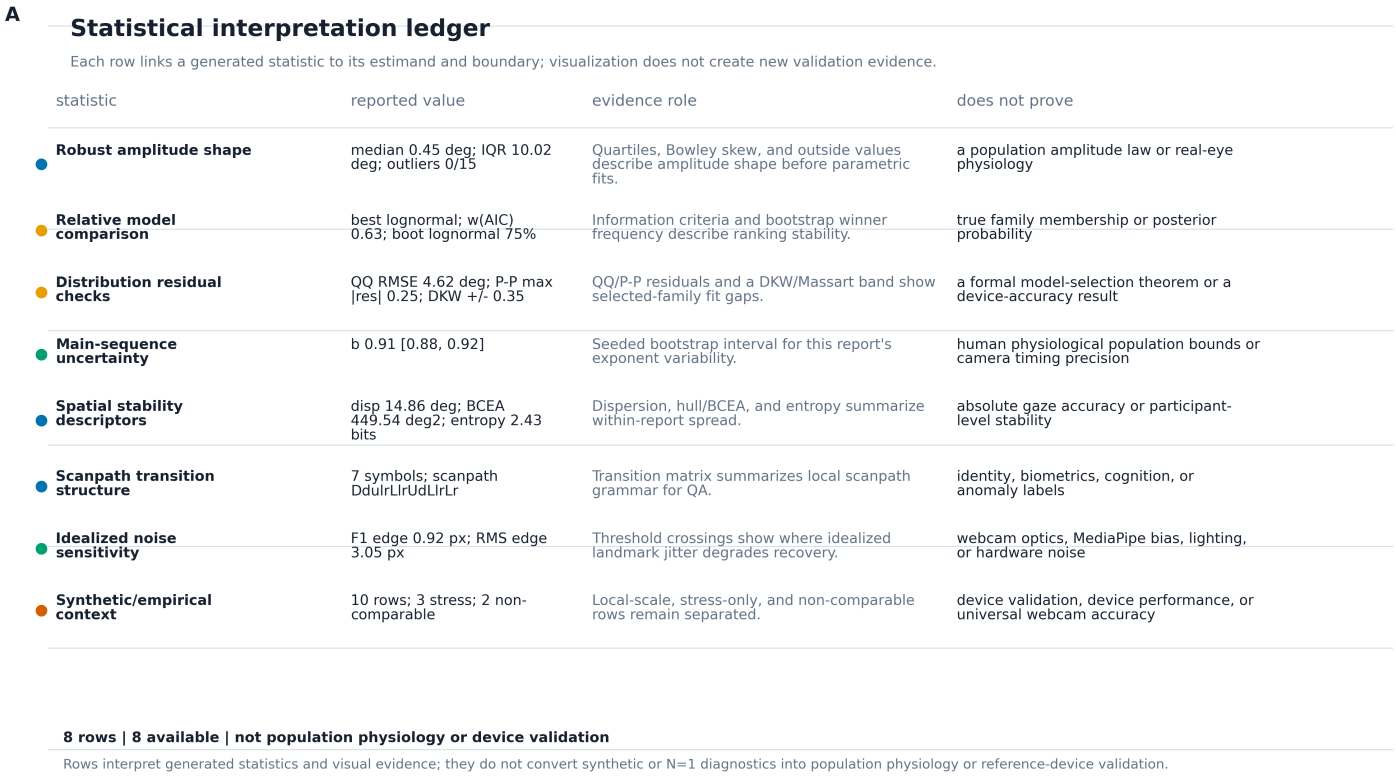


Figure 11: Statistical interpretation ledger generated from `output/figures/statistical_interpretation_ledger.json`. The rows map robust shape summaries, relative distribution comparison, CDF residual checks, bootstrap main-sequence uncertainty, spatial descriptors, scanpath transitions, idealized noise sensitivity, and the synthetic-to-empirical range bridge to their source artifacts, estimands, and explicit non-claims. The figure improves statistical readability without adding new evidence; it states what each statistic does not prove, including population physiology, posterior model truth, real-eye accuracy, and device validation.

the numbers below are internal-consistency checks on the metrics and detectors, not measurements of a real eye (sec. 23).

21.1 Self-similarity is a tautology, used as a control

Comparing a synthetic session against an exact copy of itself returns a similarity of 1.0, and comparing it against a deterministically time-shifted copy returns the same value once the paths are aligned. This is true *by construction* — identical fixation centroids and saccade sequences cannot differ — so it is reported not as a finding but as the degenerate anchor every similarity measure must hit. Its value is as a negative control: a measure that failed to return 1.0 on a session against itself would be broken, and perturbing one fixation by a known offset drops the similarity below 1.0 in the expected direction, confirming the measure responds to real differences rather than always saturating.

21.2 Adaptive and fixed thresholds recover the same scripted events

The scripted multi-saccade sessions are built so that fixed-threshold I-VT (Salvucci and Goldberg 2000) and a data-adaptive threshold derived from the velocity distribution (Nyström and Holmqvist 2010) should classify the same intervals. They do: on the synthetic grid both recover the same saccade count with onsets agreeing to within a sample, and the adaptive threshold settles inside the band of fixed thresholds that bracket the clean velocity gap. Where the signal carries injected noise the two diverge first on the smallest events, the same fragility that the saccade-F1 collapse of sec. 17 makes quantitative. The point is a cross-check between two independent detector designs on a shared, known-truth signal — the kind of agreement an event-classification reference implementation is meant to provide (Dar et al. 2021) — not a claim that either threshold is correct for real webcam data.

21.3 Windowed event rates expose temporal structure

Sliding a fixed-width window across a session and counting events per window yields a saccade-rate and fixation-rate time series whose envelope tracks the scripted event schedule: rate peaks line up with the constructed saccade bursts and fall to zero across the long fixations, and integrating the windowed counts recovers the total event counts of the whole-session report. The microsaccade rate over a fixational-jitter session stays near zero except around the single injected microsaccade, the temporal counterpart to the amplitude separation seen in sec. 14. These windowed rates are descriptive summaries of the detected event stream; on a synthetic session they recover the schedule we wrote in, and on a real recording they would inherit exactly the detector fragilities catalogued in sec. 17, with no additional accuracy implied.

Taken together, the spatial spread metrics (dispersion, convex-hull area, BCEA, and the two scanpath entropies of sec. 11) and the timeline views give a consistent, reproducible picture of the synthetic sessions: structured scanpaths read as low direction-transition entropy and a compact hull, diffuse ones as the opposite, and the temporal rates localise each event in time. Every number is a property of the detected events on a known-truth signal — a verification that the metrics and detectors behave as specified, kept strictly separate from any claim about real-eye behaviour.

22 Discussion

22.1 What the decoupling buys

By implementing established algorithms in a hardware-free core and verifying them against constructed ground truth, iTrace makes the scientific layer continuously testable and reproducible. A regression in the saccade detector, the microsaccade threshold, or the blink interpolation is caught by a headless test run, not discovered later on real data. The optional capture shell can then evolve with new landmark backends, browser capture, or deep-learning gaze behind the same typed stream boundary. This reduces coupling but does not eliminate risk: new inputs can violate numerical or scientific assumptions. The local HTML orchestrator is an example of this discipline: the browser receives eye crops and plot-ready summaries, but all capture, estimation, event detection, statistics, and export remain in the same Python package path tested elsewhere. This is the central design message of fig. 1: the camera path is an input shell, while the evidence path is a reproducible sequence of typed records, tested algorithms, and rendered reports.

22.2 What the expanded analysis layer adds

The same decoupling extends past detection into description. On top of the estimators sits a quantitative layer (`itrace.stats`) and a visualization layer (`itrace.viz`), both pure-core in spirit: the statistics modules import only NumPy/SciPy, and matplotlib is confined to the optional `figures` extra so the core stays headless. The statistics layer turns the typed events of sec. 5 and sec. 6 into descriptive summaries of fixation and saccade quantities, maximum-likelihood fits of positive-support distributions (gamma, log-normal, ex-Gaussian) ranked by AIC/BIC with a Kolmogorov–Smirnov distance, scanpath spread and organisation metrics (gaze dispersion, convex-hull area, BCEA, and spatial / direction-transition entropy), and a bootstrap-percentile confidence interval that attaches sampling uncertainty to the main-sequence exponent (sec. 11). The visualization layer renders the standard velocity, scanpath, distribution, main-sequence, microsaccade, and pupil figures — and a composite session dashboard — deterministically and byte-for-byte from a `SessionReport` (sec. 20).

The honest framing matters here too. Every one of these methods *describes* a scanpath it is handed; none of them measures real-eye accuracy or repairs the device validation gap of sec. 23. A tighter bootstrap interval, a lower KS distance, or a smaller BCEA is a property of the *detected events*, not evidence that those events match a real eye. What the layer genuinely adds is reach and reproducibility: the same descriptive vocabulary other eye-movement packages expose, computed by tested, seed-pinned functions

over the verified core, applicable unchanged to a synthetic stream with known ground truth, a real recording pushed through the pipeline, or a user-supplied benchmark file. The new `pupilseg` and `benchmark` modules are deliberately bounded examples of the same principle: the former turns a supplied eye crop into pixels or relative units without claiming millimetres; the latter compares events against supplied truth without claiming that another detector is truth. The most useful comparison point is therefore not "is iTrace more accurate than WebGazer, L2CS-Net, REMoDNaV, pymovements, PupEyes, pypillometry, ElSe, PuRe, or PupilEXT?" but "does iTrace make the exact algorithmic contract clear enough that those systems can be benchmarked against it on shared fixtures or public datasets?" (Papoutsaki et al. 2016; Abdelrahman et al. 2022; Dar et al. 2021; Krakowczyk et al. 2023; Zhang and Jonides 2026; Mittner 2020; Fuhl et al. 2015; Santini et al. 2018; Zandi et al. 2021). The guided empirical workflow sits in the same category: it makes a local session auditable by writing derived records and a summary report, while leaving the stronger reference-device study for future data.

The N=1 pilot in sec. 19 is useful precisely because it sits between the two evidence types. It does not make the real-world claims that only a reference-device study could support, but it prevents the synthetic validation from floating without an empirical scale. The synthetic sections answer whether the code recovers known signals and how idealised landmark noise degrades those signals; the pilot shows what one ordinary live run looked like in sampling rate, sampling regularity, drift, prompted-target residuals, and target acquisition latency. Those numbers are therefore best read as context for expectations and future benchmarks, not as population estimates. In practical terms, tbl. 2 and fig. 9 separate which synthetic defaults are contextualized by a local demonstration from which remain deliberate stress ranges: clean finite capture and webcam-rate timing look plausible for the demonstration path, whereas dropout, drift, landmark-noise, and calibration-bias ranges remain modelling levers that must be tested against future reference data. The statistical ledger in fig. 11 applies the same discipline to the rest of the results: relative model weights, residual diagnostics, bootstrap intervals, spatial descriptors, and scanpath summaries are reported as finite-record diagnostics with explicit non-claims.

The repeated-session ledger turns that point from prose into a collection rule. For a first paper release, additional local replicates should be treated as coverage of the declared single-participant/single-device operating scale: different lighting, display, posture, and task conditions can reveal whether the live workflow repeatedly produces finite streams, stable sampling, bounded drift, and interpretable prompted-target residuals. They should not be used to rename the study as device validation. The release gate is therefore deliberately two-part: enough repeated sessions and conditions to support the within-setup diagnostic claim, and at least one reference-device, public-dataset, or manual-annotation lane before the manuscript can claim device-level agreement. That separation keeps the v1 paper publishable as an algorithmic and workflow paper while preventing a local webcam repeatability series from becoming an unstated accuracy claim.

22.3 Algorithmic verification, not device validation

A precise claim matters. What sec. 14 establishes is *algorithmic verification*: each estimator recovers the parameters of signals produced by a known forward model. This is not *device validation* — agreement with an independent reference measurement of the same physical quantity on real eyes. We have not performed the latter. A passing synthetic suite supplies regression evidence for the tested estimation cases; by itself it is compatible with poor real-world performance. The hardware capture path is not exercised by the suite, and it is exactly where the dominant error sources live: head-pose and calibration confounds in gaze, the pupillary light reflex in pupillometry, and saccade undersampling at about 30 Hz. Until a head-to-head against a reference tracker or a public dataset exists, iTrace should be cited as a verified reference implementation of the algorithms, not a validated tracker.

22.4 The closed loop bounds, but does not establish, real error

The 3-D closed loop (sec. 16) exercises the landmark→gaze projection, but its interpretation is conditional on what the forward model and estimator *share*. Both assume an ideal pinhole projection with no lens distortion, no corneal refraction (the apparent entrance pupil and physical pupil are different optical quantities), a perfectly spherical eyeball, and a zero kappa angle. Errors from these shared idealisations are absent from the loop by construction, and the synthetic landmarks carry none of MediaPipe's real noise, bias, or eccentric-gaze foreshortening. The 0.16° residual therefore has no established ordering relative to real-world error. The $r = 1.0$ pupil result is a consistency check: correlation can remain high despite offsets, rescaling, or systematic measurement bias. Agreement requires a reference measurement and an analysis of differences in stated units (Bland and Altman 1986). Repeated observations within an individual require a design that preserves that dependence (Bland and Altman 2007). The limits are collected in sec. 23.

23 Limitations and accuracy bounds

The conveniences of the pure core do not change the physics of consumer hardware, and several limits should temper interpretation. The headline limitation is the unclosed **device validation gap**: iTrace's correctness claims rest on constructed ground truth and an internal-consistency loop, never on agreement with a reference measurement of real eyes. Everything below follows from that.

Gaze and saccades. Real-frame webcam gaze accuracy depends strongly on the dataset, camera, lighting, pose, calibration, and model class; large public benchmarks report errors in physical screen units or degrees under their own protocols rather than a universal webcam constant (Krafka et al. 2016; Zhang et al. 2017; Abdelrahman et al. 2022). iTrace's noise-sensitivity result sharpens the method-level risk: sec. 17 shows overlap-based event recovery degrading at small landmark perturbations under the declared model. Overlap F1 does not establish onset/offset accuracy, and this synthetic result is not a universal webcam timing limit.

Pupil. Webcam pupil diameter now has dedicated learned-model and dataset work (Shah et al. 2025), mature pupil-detection algorithms and platforms (Fuhl et al. 2015; Santini et al. 2018; Zandi et al. 2021), and downstream packages for transparent preprocessing, visualization, and statistical modelling once pupil samples exist (Zhang and Jonides 2026; Mittner 2020). iTrace now includes a pure eye-crop segmentation helper, but it reports image pixels or pupil/iris-relative values only. It does not implement an absolute millimetre calibration path, and its live MediaPipe pupil channel remains a relative proxy. The package surfaces this rather than implying IR-grade precision, and the pupil noise-sensitivity result is conditional on a modelling assumption (sec. 10), not measured. Gaze-position-dependent pupil foreshortening is another confound demonstrated in controlled artificial-eye experiments (Hayes and Petrov 2016); iTrace does not claim to implement or validate a correction for it.

The expanded analysis layer describes, it does not validate. The descriptive statistics, distribution fits, scanpath spread and entropy metrics, bootstrap confidence interval of sec. 11, and external benchmark report all operate on the events or truth files they are handed. They quantify the *shape* of a scanpath and the agreement with a supplied label set precisely, but a crisp gamma fit, a narrow exponent interval, a small BCEA, or high agreement with a comparator detector says nothing about whether the underlying gaze tracked a real eye — it is description downstream of the same unvalidated estimates or caller-supplied labels, not independent corroboration of them.

The live HTML app is a diagnostic interface, not a validation study. A large zoomed eye crop, stable FPS counter, and plausible rolling saccade/pupil plots prove that the webcam, MediaPipe, and Python analysis path are wired together. They do not prove gaze accuracy, pupil-diameter accuracy, or event timing agreement with an independent reference device. The guided empirical-session workflow improves that diagnostic layer by recording fixed-gaze, reading, and center/four-corner target trials and reporting drift, jitter, finite fractions, held-out target residuals, and target acquisition latency. Those quantities are valuable for the current session and screen geometry, but the prompts are not a reference eye tracker; they do not establish accuracy across cameras, participants, lighting, head pose, or tasks.

Prompted sessions support diagnostic release, not device validation. The empirical-session manifest and summary figure provide a disciplined way to add repeated sessions from the planned single participant and single device. The five collected sessions are enough for the declared diagnostic-v1 scope because they exercise the guided workflow across the two recorded lighting conditions and preserve derived records for rerendering. They do not sample population variability, cross-device variability, or independent gaze truth. The stronger validation scope therefore remains future work: more sessions, the backlit condition, and at least one reference-device, public-dataset, or manual-annotation lane under the preregistered split/metric policy.

Idealizations the validation cannot see. The closed loop and the noise model share assumptions — pinhole optics, no refraction, spherical eye, zero kappa, iid landmark noise — and are blind to any error those introduce (sec. 22). Spatial/temporal landmark-noise correlation, heteroscedasticity, and systematic bias are unmodelled.

23.1 Future work

The honest path to closing the device validation gap is empirical: a head-to-head against a reference eye-tracker or a public webcam dataset (MPIIGaze, GazeCapture) on real frames. Gaze evaluation should report error in degrees under a declared participant/session split; GazeCapture and MPIIGaze provide gaze labels, not automatically synchronized pupil-size or physiological saccade-onset truth. For calibrated pupil measurements in common units, evaluate paired differences, bias, absolute errors, and difference-versus-mean plots against a prespecified acceptable disagreement. Limits of agreement require an appropriate repeated- measurement design, and correlation alone is insufficient (Bland and Altman 1986, 2007). A relative proxy must first have a declared calibration or normalization target before comparison with millimetres. The guided session protocol added here should become a preregistered calibration and screen-target block inside that larger study, not a substitute for it. Beyond that, a calibrated absolute-diameter pupil model trained on PupilSense/EyeDentify data (Shah et al. 2025) or validated against PupilEXT-style measurement protocols (Zandi et al. 2021), smooth-pursuit and post-saccadic-oscillation classification to match REMoDNaV's four-class output (Dar et al. 2021), screen-space calibration, and binocular disparity each slot in behind the existing typed interfaces without disturbing the verified core.

24 Conclusion

iTrace shows that auditable webcam eye-movement analysis follows from a single architectural commitment: keep the verifiable algorithms pure and verify them against ground truth, and keep the fragile hardware in a thin, optional shell. The separation makes the numerical contracts testable. On the verified core sits an analysis surface — descriptive event statistics, distribution fitting and comparison, scanpath spread and entropy metrics, a bootstrap interval on the main-sequence exponent, benchmark scoring for supplied truth files, bounded live empirical diagnostics, and a deterministic figure gallery — that extends the package's descriptive reach without weakening any correctness claim. Each layer is tested, seed-pinned, and explicit about the truth source it was handed.

The contribution is honestly scoped: a verified, type-safe, reproducible reference implementation of gaze, saccade, pupil, and scanpath analysis algorithms, with a 3-D-forward-model internal-consistency check and an idealized noise-sensitivity analysis showing that the tested velocity-thresholded event recovery is sensitive to small landmark perturbations under the declared noise model. Statistical summaries expose resampling assumptions, failed draws, and the distinction between descriptive fit and inferential evidence. What remains is the empirical step that this paper deliberately does not claim to have completed: real frames, a reference device or public dataset, and protocol-specific estimates of error and agreement with uncertainty.

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26 Supplement S1: statistical methods

This supplement gives the full statistical detail behind sec. 17 so the analysis is reproducible and auditable.

26.1 Estimator and interval

For each (noise level, metric) we collect n independent trial values $\{x_i\}$ and report the sample mean $\bar{x}$ with a two-sided 95% **bootstrap percentile** confidence interval: we draw $B = 2000$ resamples of size n with replacement, take the mean of each, and read the 2.5th and 97.5th percentiles of the resampled means.

We deliberately avoid a symmetric Student-t interval `xbar +/- t_(0.975,n-1) * s / sqrt(n)` here because two of the three metrics are **bounded** — saccade F1 $\in [0, 1]$ and pupil correlation $\in [-1, 1]$ — and at low noise both sit near 1.0, exactly where a symmetric interval can extend past the bound and report a nonsensical value above 1. A percentile of resampled means cannot leave the observed bounds. This support property does not establish 95% coverage: for example, an all-success Bernoulli sample produces the interval $[1, 1]$ even when the generating probability is below one. These are approximate, pointwise intervals; coverage for the sweep settings has not been established (Efron 1987; Morris et al. 2019).

26.2 What the interval means (and the determinism caveat)

The pipeline is fully deterministic: re-running it reproduces every number byte-for-byte (sec. 18). The confidence interval therefore does **not** describe "what happens if I rerun the code." It describes the variability of the mean **across the random landmark-noise realisations sampled by the n seeded trials** under the declared scene and noise process. This Monte Carlo uncertainty interval is distinct from the standard error of the mean, $sn^{-1/2}$ for trial standard deviation s, reported separately in the sidecar (Morris et al. 2019). It estimates neither between-participant variability nor uncertainty in how well the noise model represents a real camera.

26.3 Trial independence and determinism

Each trial uses a distinct deterministic seed (`base_seed + level_index*1000 + trial`), and the bootstrap resampler is itself fixed-seeded. Pseudorandom draws are treated as independent under the generator design; distinct integer seeds are a reproducibility policy, not a proof of independence. Reproduction is conditional on the recorded runtime and locked numerical dependencies.

26.4 Saccade detection scoring

Saccade F1 uses **interval-overlap** matching against the ground-truth saccade intervals using the same greedy one-to-one matcher as external benchmark scoring. Predictions are processed in onset order and each selects the unmatched truth interval with the largest strictly positive overlap. Touching endpoints do not match; no interval can contribute more than one true positive. Fragmentation and merging are therefore penalised. Precision, recall, and $F_1 = 2PR/(P + R)$ follow from these counts. This greedy rule is deterministic, not a globally optimal assignment. Even one-to-one overlap F1 does not establish onset/offset accuracy: a greatly extended interval can still overlap a true event. Boundary errors require separate matched-event timing metrics.

This revision replaces the archived release's many-to-many any-overlap sweep scorer. The regenerated sweep and thresholds belong to the revised scorer; historical release artifacts retain their original interpretation.

26.5 Threshold crossings

A usability threshold (gaze RMS = 2°, saccade F1 = 0.8, pupil $r = 0.9$) is located by linear interpolation between the two grid points that bracket the crossing; with $n = 25$ these crossings are approximate and should be read to ~2 significant figures. The crossing in physical units uses `sigma_to_pixels(σ, image_width)` (sec. 10).

26.6 One source of truth

The figure (fig. 6) and the table (tbl. 1) are both rendered from a single `NoiseSweepResult` via `power.summary_records` / `format_summary_markdown`, so prose, plot, and table cannot drift apart — verified by a cell-for-cell match in the documentation audit.

27 Supplement S2: software architecture

27.1 Module map

Module	Layer	Responsibility
`config`	core	frozen configs for detection, pupil, capture, figures
`types`	core	dataclasses + units (`GazeStream`, `PupilStream`, events)
`geometry`	core	pix↔deg, iris→angle, interocular normalisation
`calibration`	core	affine gaze calibration and recording-quality summaries
`velocity`	core	Savitzky–Golay / gradient velocity
`saccades`	core	I-VT, I-DT, Engbert–Kliegl microsaccades
`detection`	core	adaptive I-VT, PSO candidates, ISI, acceleration
`mainsequence`	core	saturating + power-law fit
`pupil`	core	deblink, MAD, low-pass, baseline
`pupilseg`	core	deterministic eye-crop segmentation; pixels or relative units
`pupilphase`	core	causal phase detector
`encoding`	core	direction-character + n-gram scanpath
`eyemodel`	core	3-D eyeball + pinhole projection
`scene`	core	animated trajectory + closed loop
`synthetic`	core	seeded gaze/pupil sessions with truth records
`power`	core	noise-sensitivity sweep + statistics
`benchmark`	core	user-supplied truth/comparator event scoring
`experiments`	core	guided derived-session protocols and empirical reports
`stats`	core	descriptive, distribution, similarity, event, bootstrap metrics
`pipeline` / `io`	core	end-to-end analysis + CSV
`capture`	shell	OpenCV + MediaPipe (lazy)
`live`	shell	local FastAPI/WebSocket HTML orchestrator (lazy)
`dashboard`	shell	Streamlit + Plotly (lazy)
`viz`	shell	matplotlib figure gallery (lazy via figures extra)
`cli`	shell	typer command-line interface

27.2 Dependency contract

The shell may import the core; the core may never import the shell. Runtime optional dependencies (`opencv-python`, `mediapipe`, `fastapi`, `uvicorn`, `streamlit`, `plotly`, `matplotlib`) are imported lazily inside functions or the optional `itrace.viz` subpackage, never by `import itrace`, and absence raises an actionable error rather than breaking import. The Starlette/FastAPI `TestClient` HTTP client dependency (`httpx2` in current releases, with `httpx` kept for older compatibility) is kept in the `dev` extra rather than the runtime `web` or `all` extras. `import itrace` and the full test suite run with none of the hardware/dashboard/web runtime dependencies installed; the `all` extra installs capture, dashboard, figure, and web backends together for local smoke testing.

27.3 Test strategy and gates

Tests are **no-mocks**: each detector is verified by synthesising a signal whose ground truth is known by construction and asserting recovery within a stated tolerance, plus an independent reference implementation for cross-checking (sec. 18). Promotion gates: `pytest` green at ≥90% statement+branch coverage, `ruff check` and `ruff format --check` clean, and `mypy --strict` clean. Figures and tables are regenerated by thin orchestrator scripts under `scripts/` that contain no analysis logic of their own. Rendered manuscript metrics are hydrated from `docs/verification_metrics.json`, preventing stale literal test counts or coverage percentages from becoming a second source of truth.

28 Supplement S3: resampling design and inferential boundaries

28.1 Estimand, independent unit, and observed record

The noise sweep estimates the expected recovery metric for a fixed synthetic scene under a specified perturbation generator. Its independent unit is an entire generated trial. Event-wise bootstraps instead perturb the events detected in a supplied recording. Neither design estimates between-person uncertainty from the single participant represented in the empirical ledgers. Repeated sessions from one person do not become independent people when resampled (Bland and Altman 2007; Morris et al. 2019).

`bootstrap_indices` supports IID indices and overlapping moving blocks of a declared number of consecutive observations. Block starts are sampled uniformly from the available complete blocks; concatenation is truncated to the original record length, without wrapping across the record boundary. `bootstrap_statistic` rejects missing or nonfinite input rather than silently deleting rows and changing adjacency. Pair resampling uses the same indices for amplitude and velocity. The default block length of one reproduces IID resampling; longer blocks require an approximately stationary sequence and a scientifically justified length (Künsch 1989). A block as long as the record has no resampling variation. Nonstationary task epochs, gaps, and participant clusters need a design tailored to those units; an arbitrary block length is not a universal correction. The controlled illustration in fig. 12 makes the effect of changing that resampling unit visible without assuming it validates coverage.

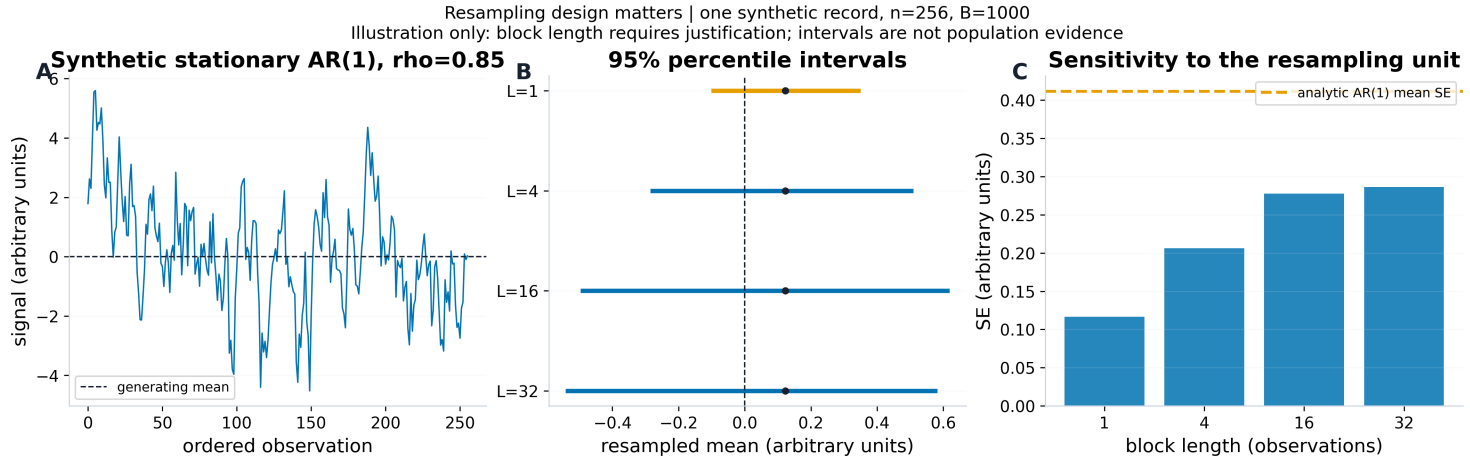


Figure 12: Resampling sensitivity for one seeded stationary Gaussian AR(1) record in arbitrary units. Panel A shows the ordered observations; panel B displays exact percentile interval endpoints for the mean under IID resampling (L=1) and several moving-block lengths; panel C compares resampled standard errors with the finite-record analytic AR(1) mean standard error. All quantities come from the accompanying JSON, with the seed and design recorded. This example illustrates sensitivity to the resampling unit; it is not eye data, a coverage experiment, or a recommendation for a particular block length.

For this illustration, innovations have variance one and the initial observation has stationary variance. With n observations and lag correlation ρ, the known generating variance of the sample mean is

$$V = (n + 2S)/(n^2(1 - \rho^2))$$

where V denotes the variance of the sample mean and `S = sum((n - h) * rho**h for h in range(1, n))`.

The analytic reference is checked against a separately constructed covariance matrix. Agreement of one interval with its generating mean is not evidence of nominal interval coverage; that would require repeated independent datasets and Monte Carlo uncertainty for the coverage estimate (Morris et al. 2019).

28.2 Failure accounting and fit calibration

Main-sequence uncertainty uses log-log fits alone, without fitting the unrelated saturating model on each draw. Draws with insufficient unique amplitudes are counted as failures. The detailed API and figure expose attempted and successful counts; an interval

based on the remaining draws is conditional on success. No successful draws produces an unavailable interval, never artificial precision. Percentile endpoints are displayed directly, even when they do not bracket the original point estimate (Efron 1987; Morris et al. 2019).

The fixed-family KS refit bootstrap generates a sample from the fitted null, refits all originally estimated parameters, and computes its fitted-CDF distance. Originally fixed parameters stay fixed. For r null distances at least as large as observed among B draws, the reported probability is $(r+1)/(B+1)$; the minimum nonzero probability is $1/(B+1)$, disclosed as Monte Carlo resolution. Every requested null fit must succeed. The publication diagnostic applies this procedure to the AIC-selected family without repeating family selection, so it remains explicitly exploratory. It is not selection-adjusted, does not establish serial independence, and does not accept a model as true. QQ, P-P, KS, and AD/CvM-style distances remain useful descriptive checks under these limits.

28.3 What would support stronger claims

Future confirmatory comparisons should prespecify the independent unit, target metric, candidate models, missingness policy, calibration split, acceptable error, and multiplicity policy. Reusing calibration observations for evaluation or selecting a favorable metric after seeing the data changes the question being answered. A complete evaluation should report failures and exclusions with their denominators, rather than only statistics from successful records (Morris et al. 2019). Device agreement needs measurements of comparable quantities and units; a perfectly correlated scaled signal can still disagree systematically (Bland and Altman 1986).